ENGINE CONTROL SYSTEM

1. General

The engine control system of the 2AZ-FE engine has the following features.

System	Outline
SFI [Sequential Multiport Fuel Injection]	● An L-type SFI system directly detects the intake air mass with a hot wire type mass air flow meter. ● The fuel injection system is a sequential multiport fuel injection system. ● Fuel injection takes two forms: Synchronous injection, which always takes place with the same timing in accordance with the basic injection duration and an additional correction based on the signals provided by the sensors. Non-synchronous injection, which takes place at the time an injection request based on the signals provided by the sensors is detected, regardless of the crankshaft position. ● Synchronous injection is further divided into group injection during a cold start, and independent injection after the engine is started.
ESA [Electronic Spark Advance]	● Ignition timing is determined by the ECM based on signals from various sensors. The ECM corrects ignition timing in response to engine knocking. ● This system selects the optimal ignition timing in accordance with the signals received from the sensors and sends the (IGT) ignition signal to the igniter.
ETCS-i [Electronic Throttle Control System-intelligent] [See page EG-49]	Optimally controls the throttle valve opening in accordance with the amount of accelerator pedal effort and the condition of the engine and the vehicle.
VVT-i [Variable Valve Timing-intelligent] [See page EG-51]	Controls the intake camshaft to an optimal valve timing in accordance with the engine condition.
Intake Manifold Runner Valve Control* [See page EG-55]	During idling while the engine is cold, this control closes the intake manifold runner valve in order to create a strong tumble airflow in the intake air. This promotes the mixture of air and fuel and improves combustion efficiency.
Fuel Pump Control [See page EG-58]	● Fuel pump operation is controlled by signals from the ECM. ● The fuel pump is stopped, when the SRS airbag is deployed in a frontal, side, and rear of side collision.
Air Conditioning Cut-off Control	By turning the air conditioning compressor ON or OFF in accordance with the engine condition, drivability is maintained.
Cooling Fan Control [See page EG-59]	Cooling fan operation is controlled by signals from the ECM based on the engine coolant temperature sensor signal and air conditioning operation.
Air Fuel Ratio Sensor and Oxygen Sensor Heater Control	Maintains the temperature of the air fuel ratio sensor or oxygen sensor at an appropriate level to increase accuracy of detection of the oxygen concentration in the exhaust gas.
Evaporative Emission Control [See page EG-60]	● The ECM controls the purge flow of evaporative emission (HC) in the canister in accordance with engine conditions. ● Approximately five hours after the engine switch has been turned OFF, the ECM operates the canister pump module to detect any evaporative emission leakage occurring between the fuel tank and the canister through changes in the fuel tank pressure.

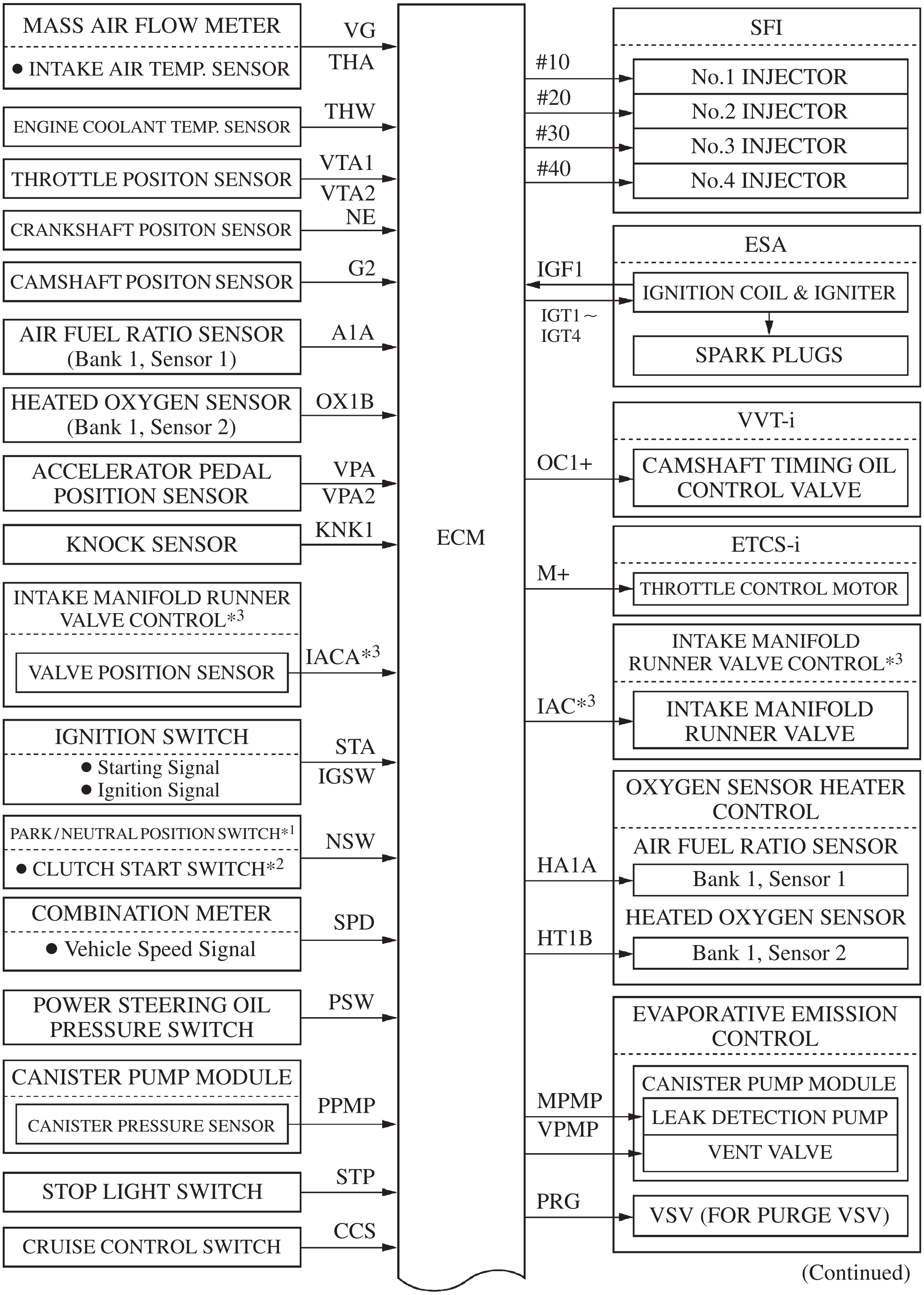
*: Only for California package models.

(Continued)

System	Outline
Engine Immobilizer	Prohibits fuel delivery and ignition if an attempt is made to start the engine with an invalid key.
Diagnosis [See page EG-72]	When the ECM detects a malfunction, the ECM diagnoses and memorizes the failed section.
Fail-Safe [See page EG-73]	When the ECM detects a malfunction, the ECM stops or controls the engine according to the data already stored in the memory.

2. Construction

The configuration of the engine control system is as shown in the following chart.



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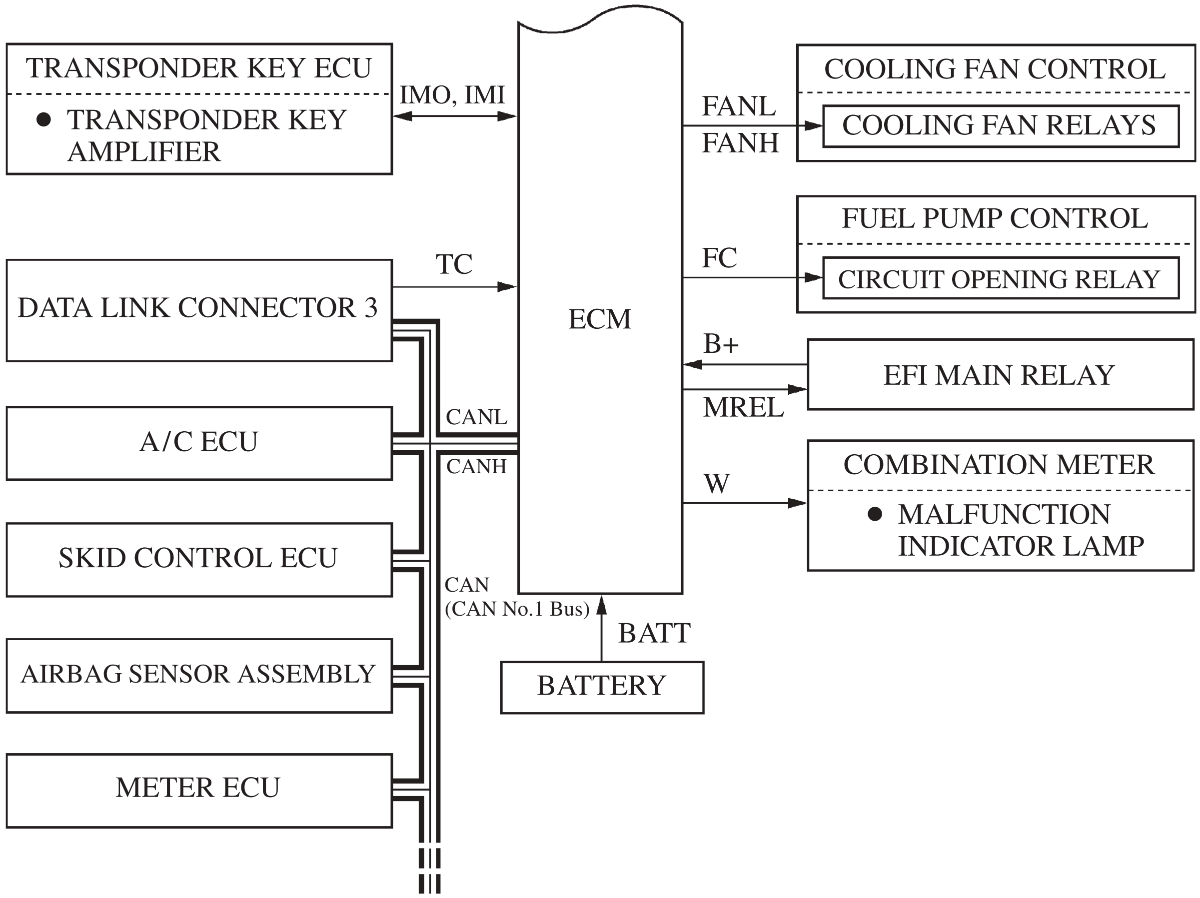

*1: With automatic transaxle model
*2: With manual transaxle model
*3: Only for California package model

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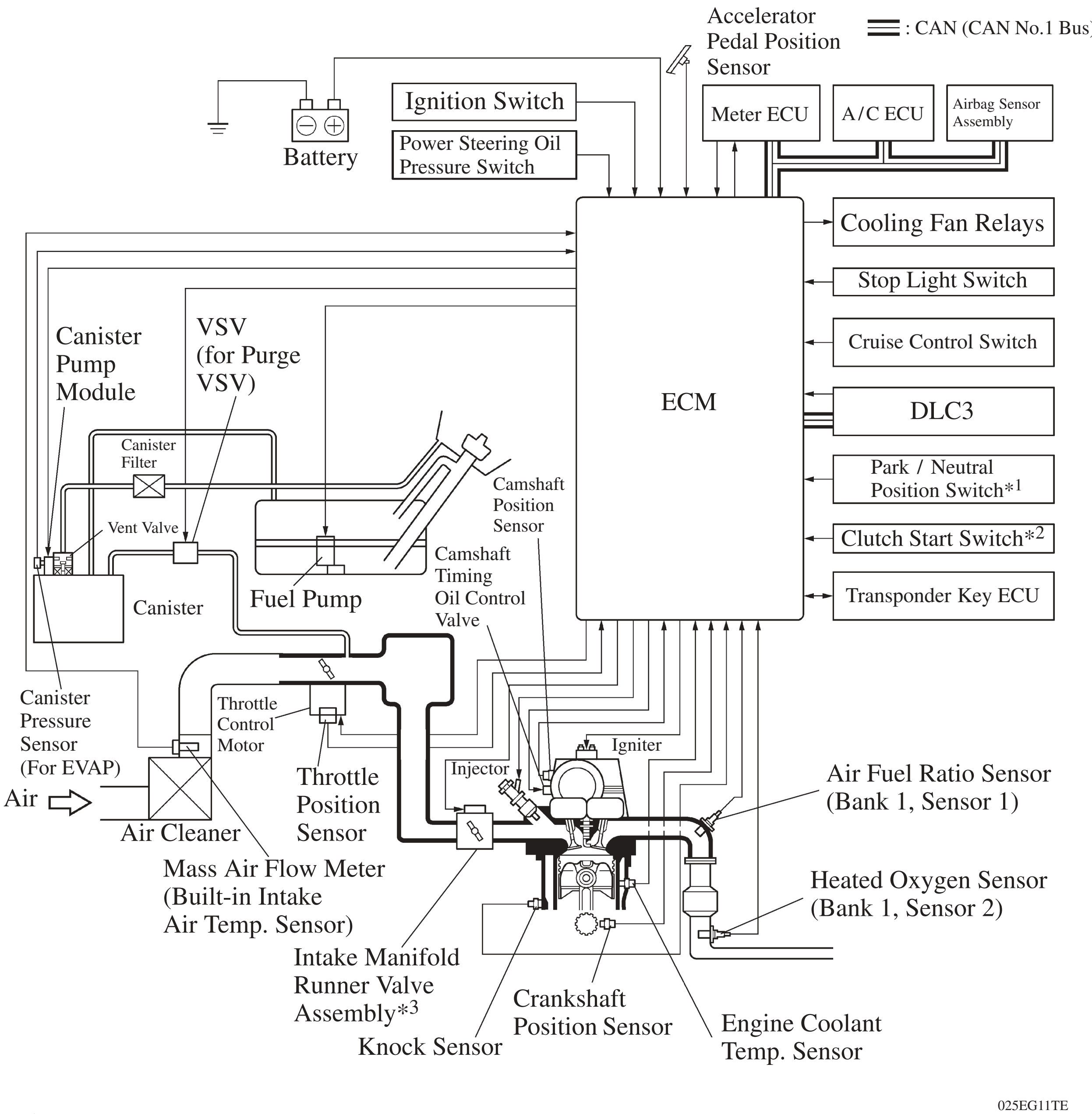
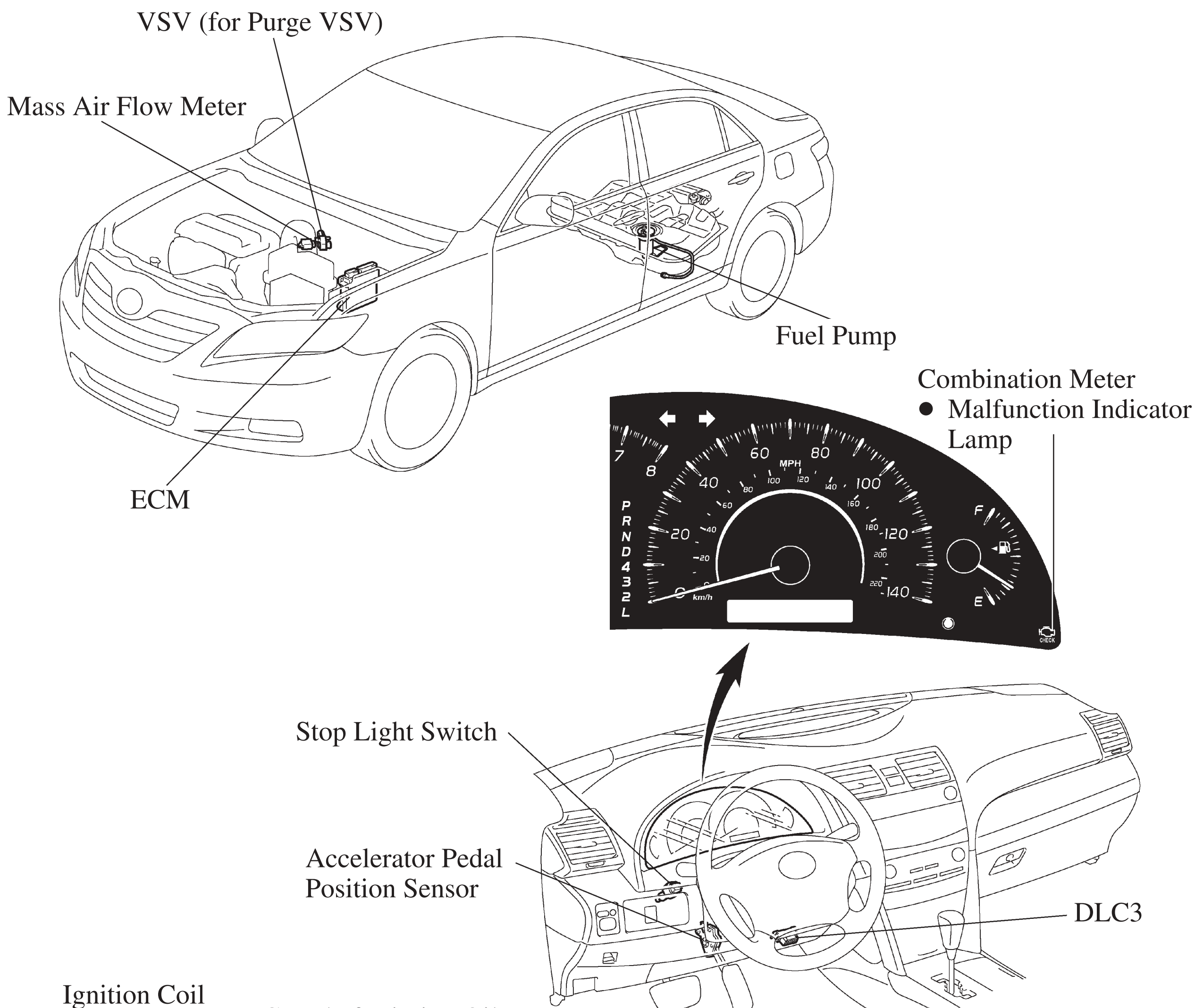
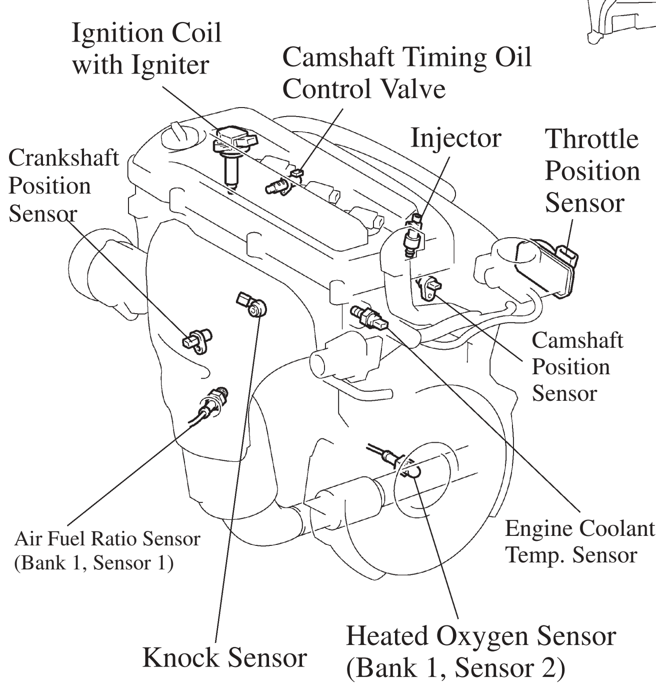
3. Engine Control System Diagram



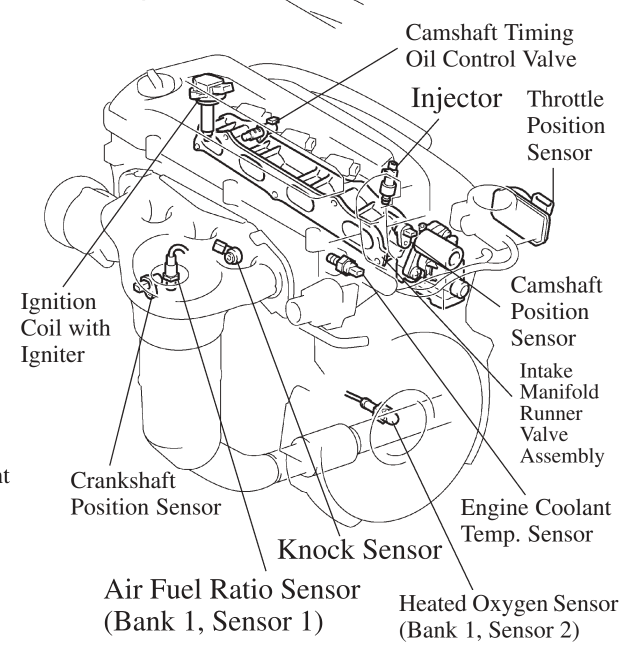
*1: With automatic transaxle model
*2: With manual transaxle model
*3: Only for California package model

4. Layout of Main Components





Except California Package Model



California Package Model

025EG12Y

5. Main Component of Engine Control System

General

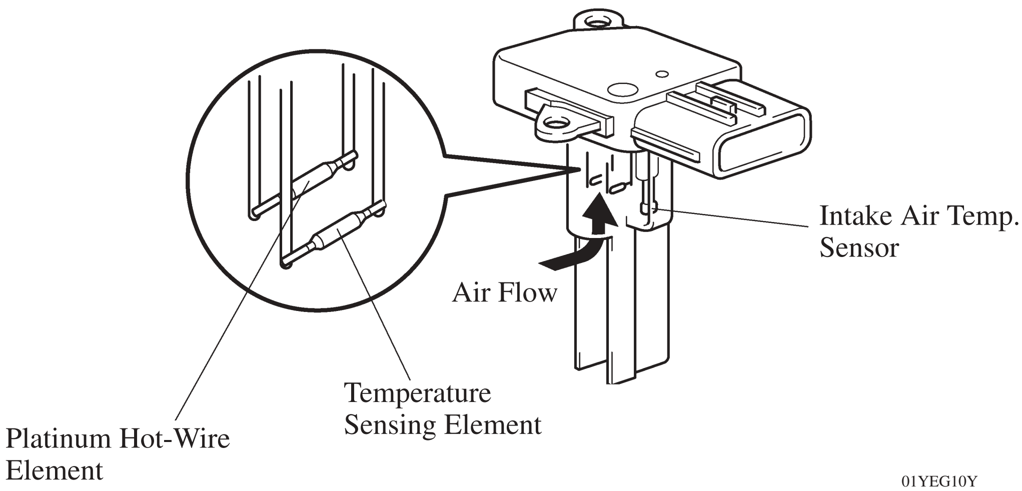
The main components of the 2AZ-FE engine control system are as follows:

Components	Outline	Quantity	Function
ECM	32-bit CPU	1	The ECM optimally controls the SFI, ESA and ISC to suit the operating conditions of the engine in accordance with the signals provided by the sensors.
Oxygen Sensor (Bank 1, Sensor 2)	Cup Type with Heater	1	This sensor detects the oxygen concentration in the exhaust emission by measuring the electromotive force which is generated in the sensor itself.
Air Fuel Ratio Sensor (Bank 1, Sensor 1)	Planar Type with Heater	1	As with the oxygen sensor, this sensor detects the oxygen concentration in the exhaust emission. However, it detects the oxygen concentration in the exhaust emission linearly.
Mass Air Flow Meter	Hot-wire Type	1	This sensor has a built-in hot-wire to directly detect the intake air mass.
Crankshaft Position Sensor (Rotor Teeth)	Pick-up Coil Type (36-2)	1	This sensor detects the engine speed and performs the cylinder identification.
Camshaft Position Sensor (Rotor Teeth)	Pick-up Coil Type (36-2)	1	This sensor performs the cylinder identification.
Engine Coolant Temperature Sensor	Thermistor Type	1	This sensor detects the engine coolant temperature by means of an internal thermistor.
Intake Air Temperature Sensor	Thermistor Type	1	This sensor detects the intake air temperature by means of an internal thermistor.
Knock Sensor	Built-in Piezoelectric Type (Flat Type)	1	This sensor detects an occurrence of the engine knocking indirectly from the vibration of the cylinder block caused by the occurrence of engine knocking.
Throttle Position Sensor	No-contact Type	1	This sensor detects the throttle valve opening angle.
Accelerator Pedal Position Sensor	No-contact Type	1	● This sensor detects the amount of pedal effort applied to the accelerator pedal. ● The sensor shape is different between TMC made models and TMMK made models. However, those are the no-contact type sensors using Hall IC.
Injector	12-Hole Type	4	The injector is an electromagnetically-operated nozzle which injects fuel in accordance with signals from the ECM.

EG

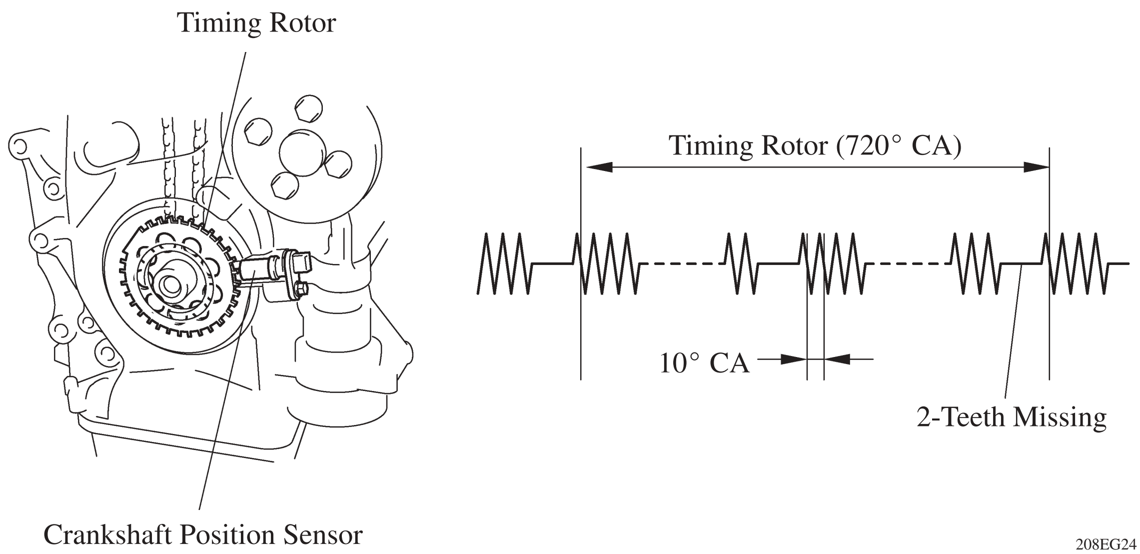
Mass Air Flow Meter

- This mass air flow meter, which is a plug-in type, allows a portion of the intake air to flow through the detection area. By directly measuring the mass and the flow rate of the intake air, the detection precision is improved and the intake air resistance is reduced.
- This mass air flow meter has a built-in intake air temperature sensor.



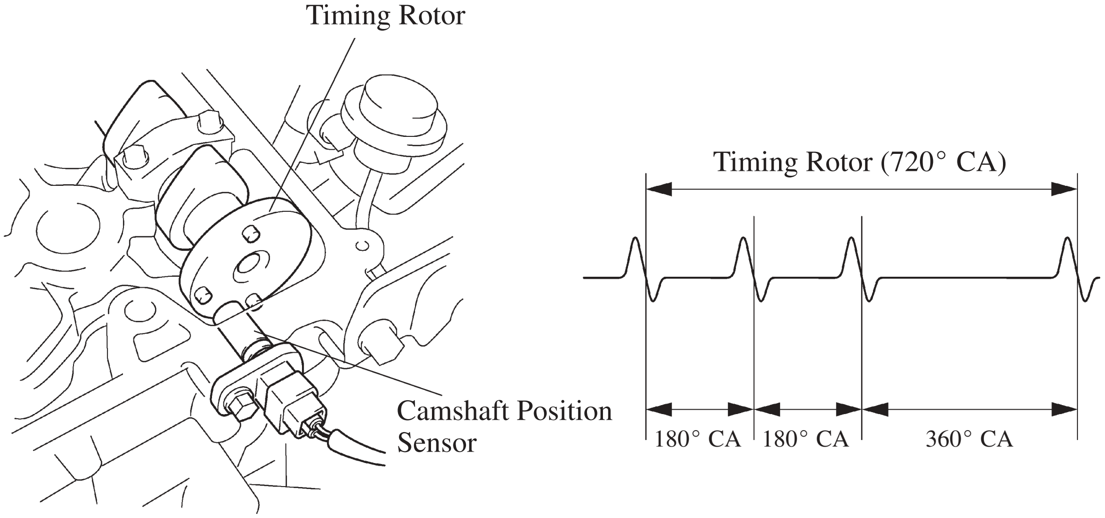
Crankshaft Position Sensor

The timing rotor of the crankshaft consists of 34 teeth, with 2 teeth missing. The crankshaft position sensor outputs the crankshaft rotation signals every 10°, and the missing teeth are used to determine the top-dead-center.



Camshaft Position Sensor

The camshaft position sensor is mounted on the left bank of cylinder head. To detect the camshaft position, a protrusion that is provided on the timing pulley is used to generate 1 pulse for every 2 revolution of the crankshaft.



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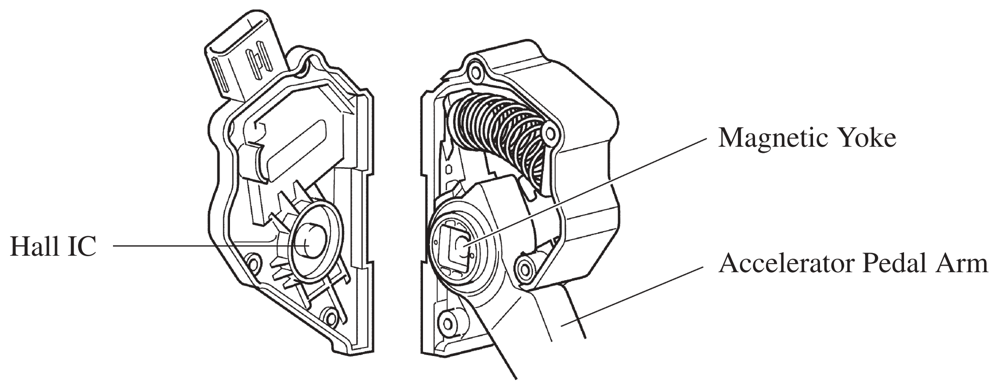
Accelerator Pedal Position Sensor

1) TMC Made Model

This no-contact type accelerator pedal position sensor uses a Hall IC, which is mounted on the accelerator pedal arm.

- The magnetic yoke is mounted at the base of the accelerator pedal arm. This yoke rotates around the Hall IC in accordance with the amount of effort that is applied to the accelerator pedal. The Hall IC converts the changes in the magnetic flux that occur into electrical signals, and outputs them in the form of accelerator pedal position signals to the ECM.
- The Hall IC contains two circuits, one for the main signal, and one for the sub signal. It converts the accelerator pedal position (angle) into electric signals that have differing characteristics and outputs them to the ECM.



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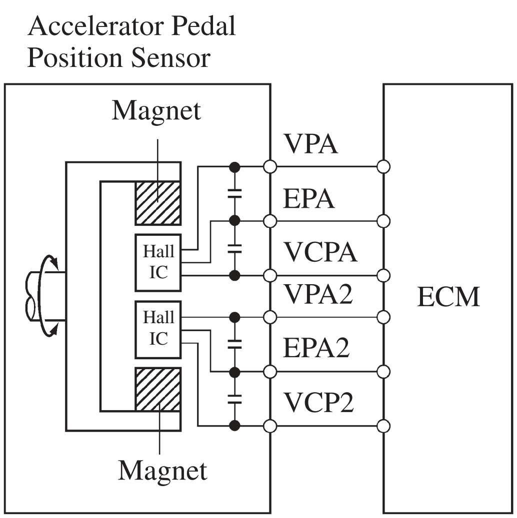

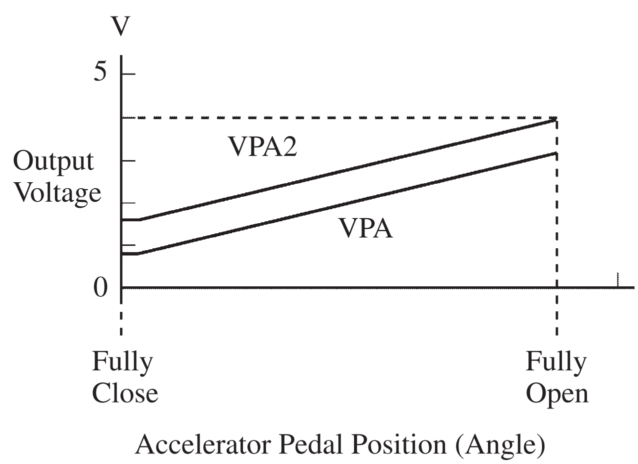

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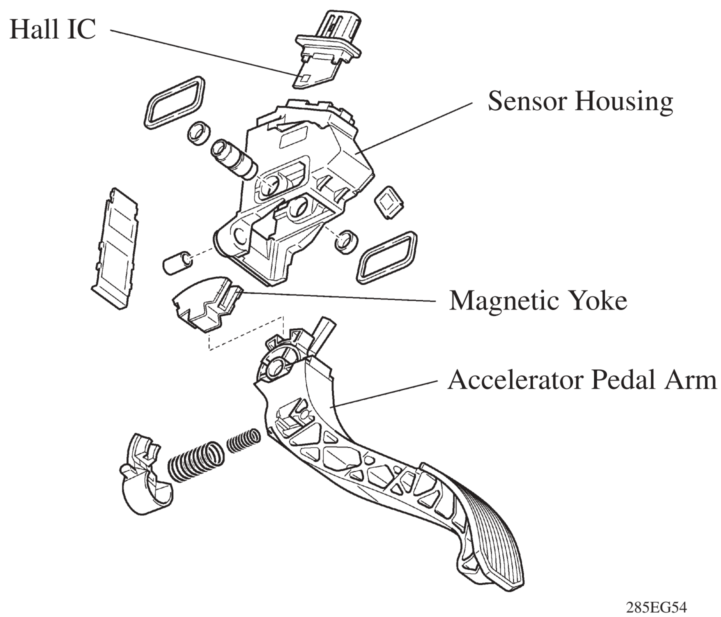
Service Tip

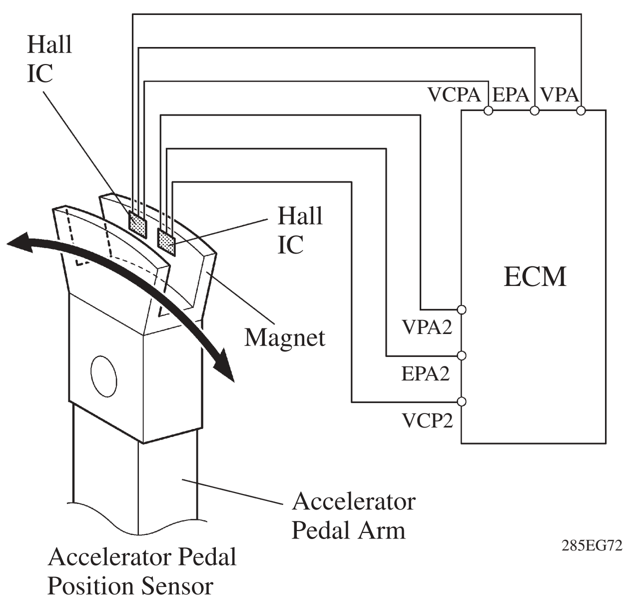
The inspection method differs from a conventional accelerator pedal position sensor because this sensor uses a Hall IC. For details, refer to the 2007 Camry Repair Manual (Pub. No. RM0250U).

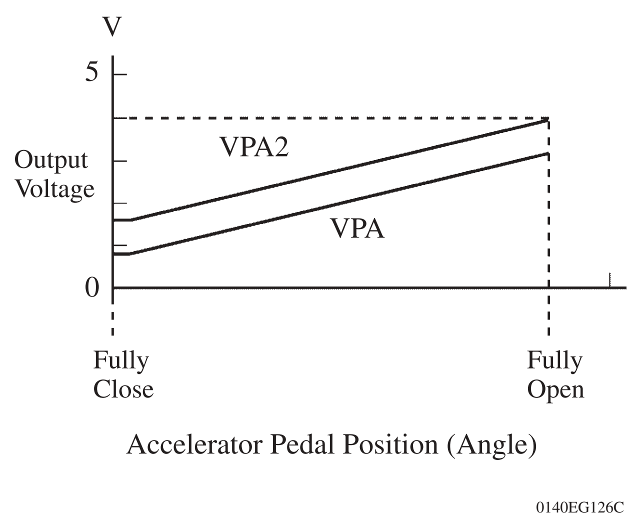
2) TMMK Made Model

The non-contact type accelerator pedal position sensor uses a Hall IC, which is mounted on the accelerator pedal arm.

- The magnetic yoke that is mounted at the base of the accelerator pedal arm moves around the Hall IC in accordance with the amount of effort that is applied to the accelerator pedal. The Hall IC converts the changes in the magnetic flux that occur at that time into electrical signals, and outputs them in the form of accelerator pedal effort to the ECM.
- This accelerator pedal position sensor includes 2 Hall ICs and circuits for the main and sub signals. It converts the accelerator pedal depressing angles into electric signals with two differing characteristics and outputs them to the ECM.





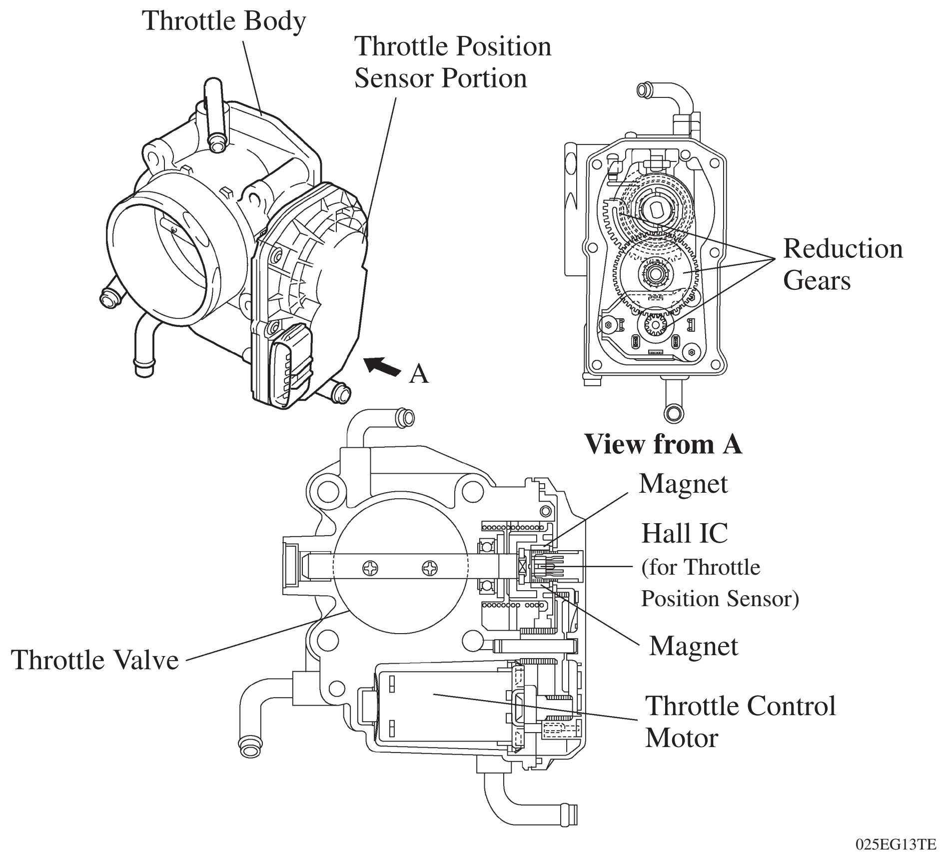


> **Service Tip**
> The inspection method differs from the conventional accelerator pedal position sensor because this sensor uses a hall IC. For details, refer to the 2007 Camry Repair Manual (Pub. No. RM0250U).

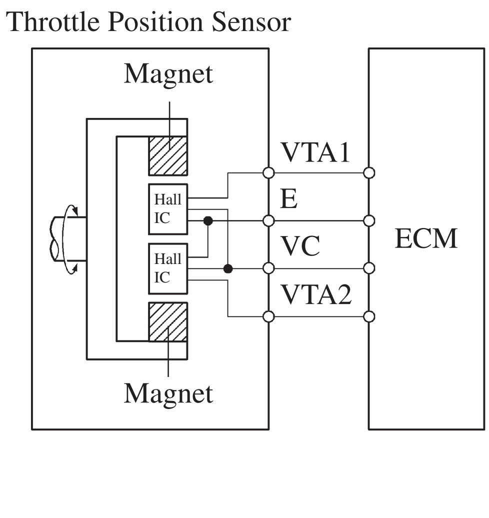
Throttle Position Sensor

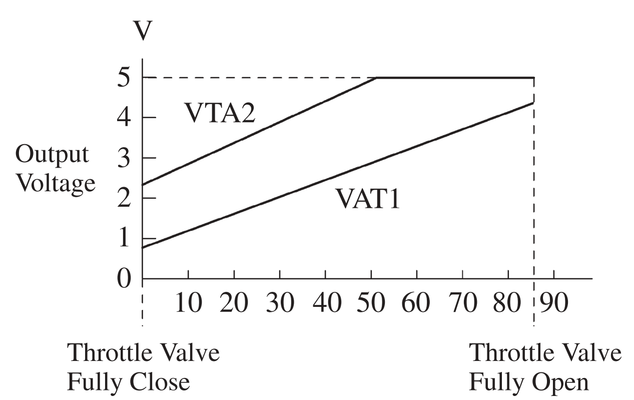
The no-contact type throttle position sensor uses a Hall IC, which is mounted on the throttle body.

- The Hall IC is surrounded by a magnetic yoke. The Hall IC converts the changes that occur in the magnetic flux at that time into electrical signals and outputs them in the form of a throttle valve effort to the ECM.
- The Hall IC contains circuits for the main and sub signals. It converts the throttle valve opening angles into electric signals with two differing characteristics and outputs them to the ECM.



Cross Section





Throttle Valve Opening Angle

Service Tip

The inspection method differs from the conventional throttle position sensor because this sensor uses a hall IC. For details, refer to the 2007 Camry Repair Manual (Pub. No. RM0250U).

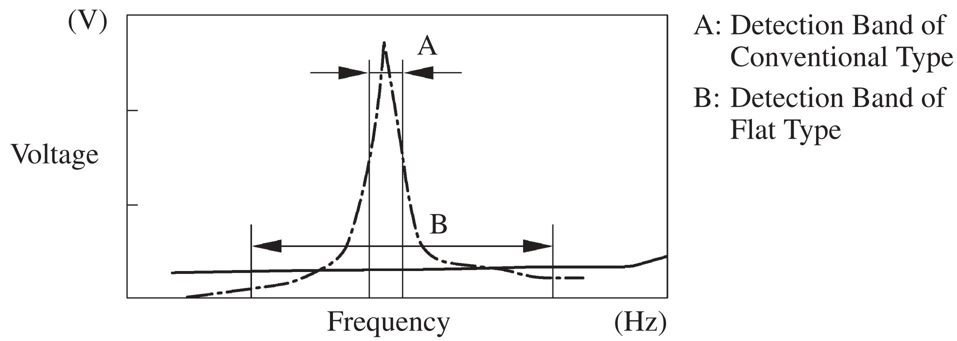
Knock Sensor (Flat Type)

1) General

In the conventional type knock sensor (resonant type), a vibration plate, which has the same resonance point as the knocking frequency of the engine, is built in and can detect the vibration in this frequency band.
On the other hand, a flat type knock sensor (non-resonant type) has the ability to detect vibration in a wider frequency band from about 6 kHz to 15 kHz, and has the following features:

- The engine knocking frequency will change a bit depending on the engine speed. The flat type knock sensor can detect vibration even when the engine knocking frequency is changed. Thus the vibration detection ability is increased compared to the conventional type knock sensor, and a more precise ignition timing control is possible.

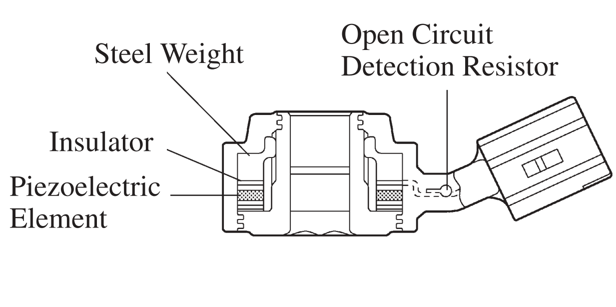
—·— : Conventional Type
——— : Flat Type



Characteristic of Knock Sensor

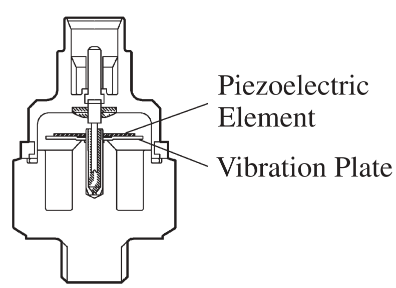
2) Construction

- The flat type knock sensor is installed on the engine through the stud bolt installed on the cylinder block. For this reason, a hole for the stud bolt is running through in the center of the sensor.
- Inside of the sensor, a steel weight is located on the upper portion and a piezoelectric element is located under the weight through the insulator.
- The open/short circuit detection resistor is integrated.



Flat Type Knock Sensor (Non-Resonant Type)

214CE01

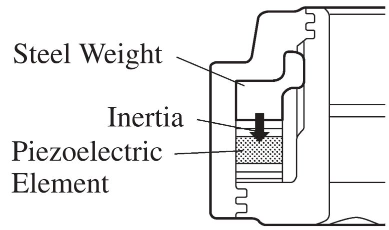


Conventional Type Knock Sensor (Resonant Type)

214CE02

3) Operation

The knocking vibration is transmitted to the steel weight and its inertia applies pressure to the piezoelectric element. The action generates electromotive force.



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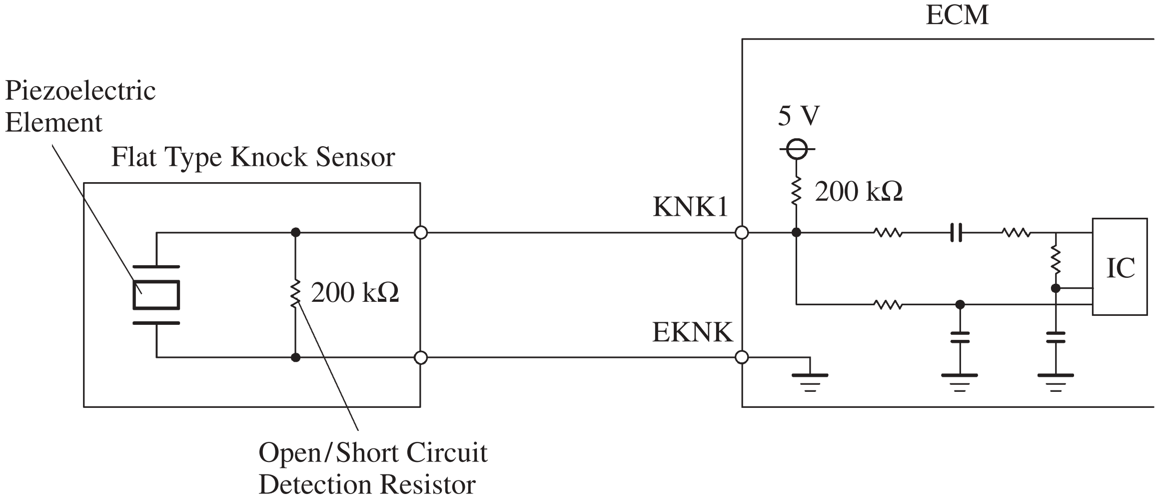
4) Open/Short Circuit Detection Resistor

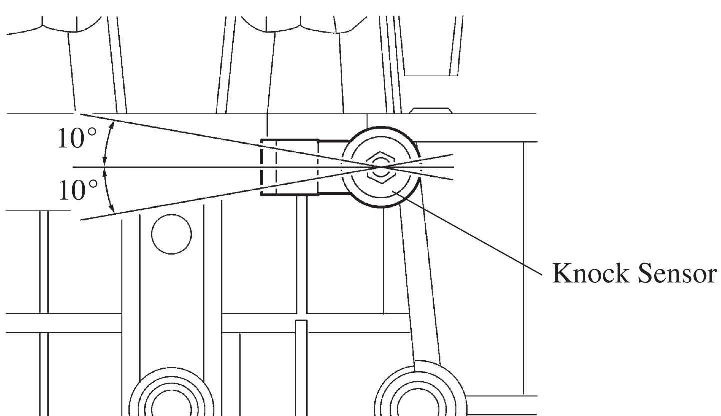
During the ignition is ON, the open/short circuit detection resistor in the knock sensor and the resistor in the ECM keep the voltage at the terminal KNK1 of engine constant.
An IC (Integrated Circuit) in the ECM is always monitoring the voltage of the terminal KNK1. If the open/short circuit occurs between the knock sensor and the ECM, the voltage of the terminal KNK1 will change and the ECM detects the open/short circuit and stores DTC (Diagnostic Trouble Code).



214CE06

Service Tip

- In accordance with the adoption of open/short circuit detection resistor, the inspection method for the sensor has been changed. For details, refer to 2007 Camry Repair Manual (Pub. No. RM0250U).
- To prevent the water accumulation in the connector, make sure to install the flat type knock sensor in the position as shown in the following illustration.



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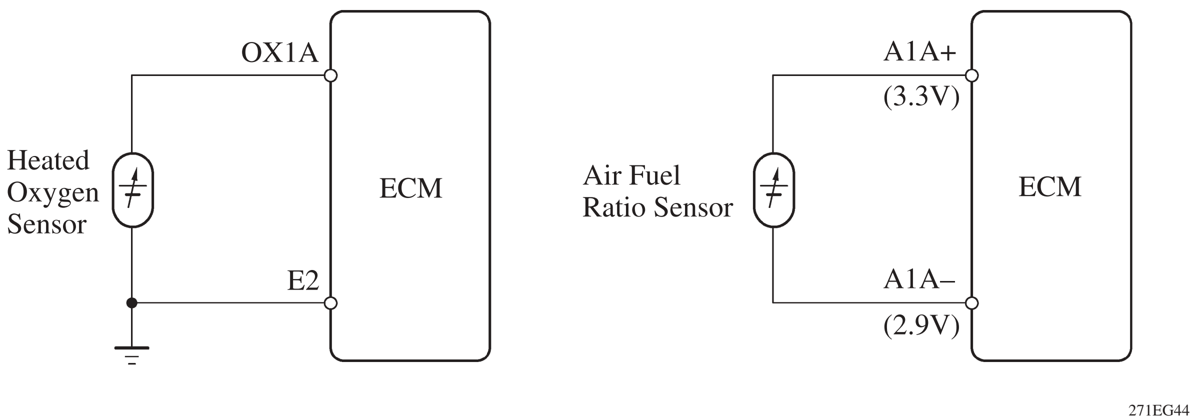
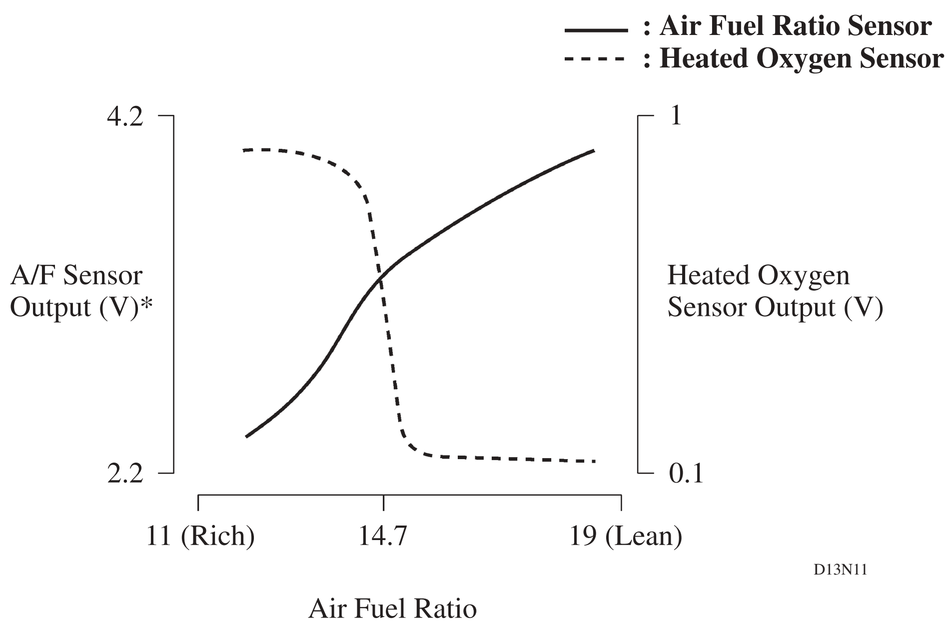
Heated Oxygen Sensor and Air Fuel Ratio Sensor

1) General

- The heated oxygen sensor and the air fuel ratio sensor differ in output characteristics.
- The output voltage of the heated oxygen sensor changes in accordance with the oxygen concentration in the exhaust gas. The ECM uses this output voltage to determine whether the present air-fuel ratio is richer or leaner than the stoichiometric air-fuel ratio.
- Approximately 0.4 V is constantly applied to the air-fuel ratio sensor, which outputs an amperage that varies in accordance with the oxygen concentration in the exhaust gas. The ECM converts the changes in the output amperage into voltage in order to linearly detect the present air-fuel ratio.



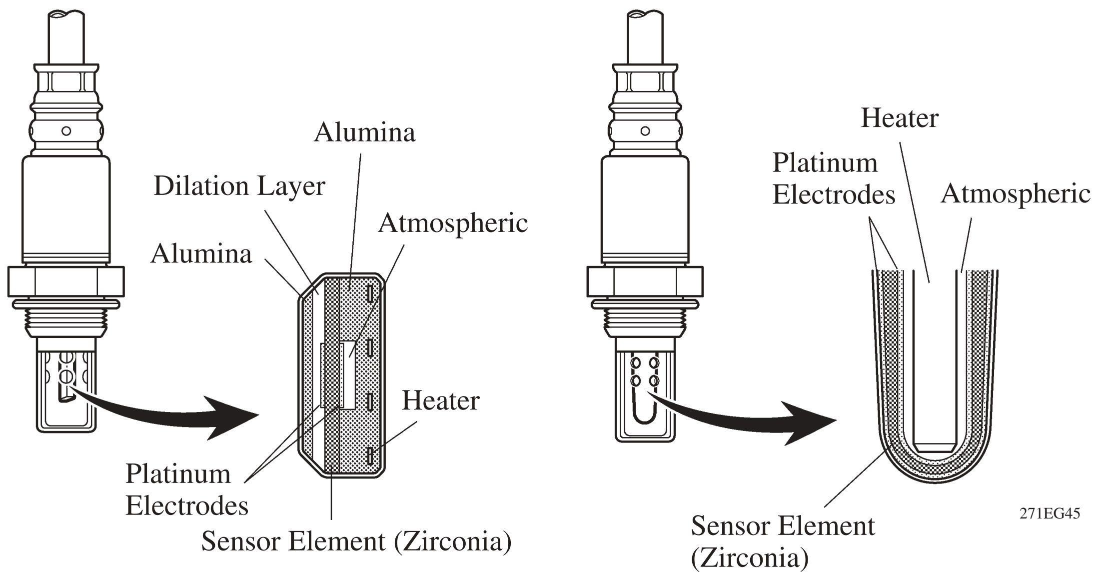
Heated Oxygen Sensor Circuit **Air Fuel Ratio Sensor Circuit**



*: This calculation value is used internally in the ECM, and is not an ECM terminal voltage.

2) Construction

- The basic construction of the heated oxygen sensor and the air-fuel ratio sensor is the same. However, they are divided into the cup type and the planar type, according to the different types of heater construction that are used.
- The cup type sensor contains a sensor element that surrounds a heater.
- The planar type sensor uses alumina, which excels in heat conductivity and insulation, to integrate a sensor element with a heater, thus improving the warm-up performance of the sensor.



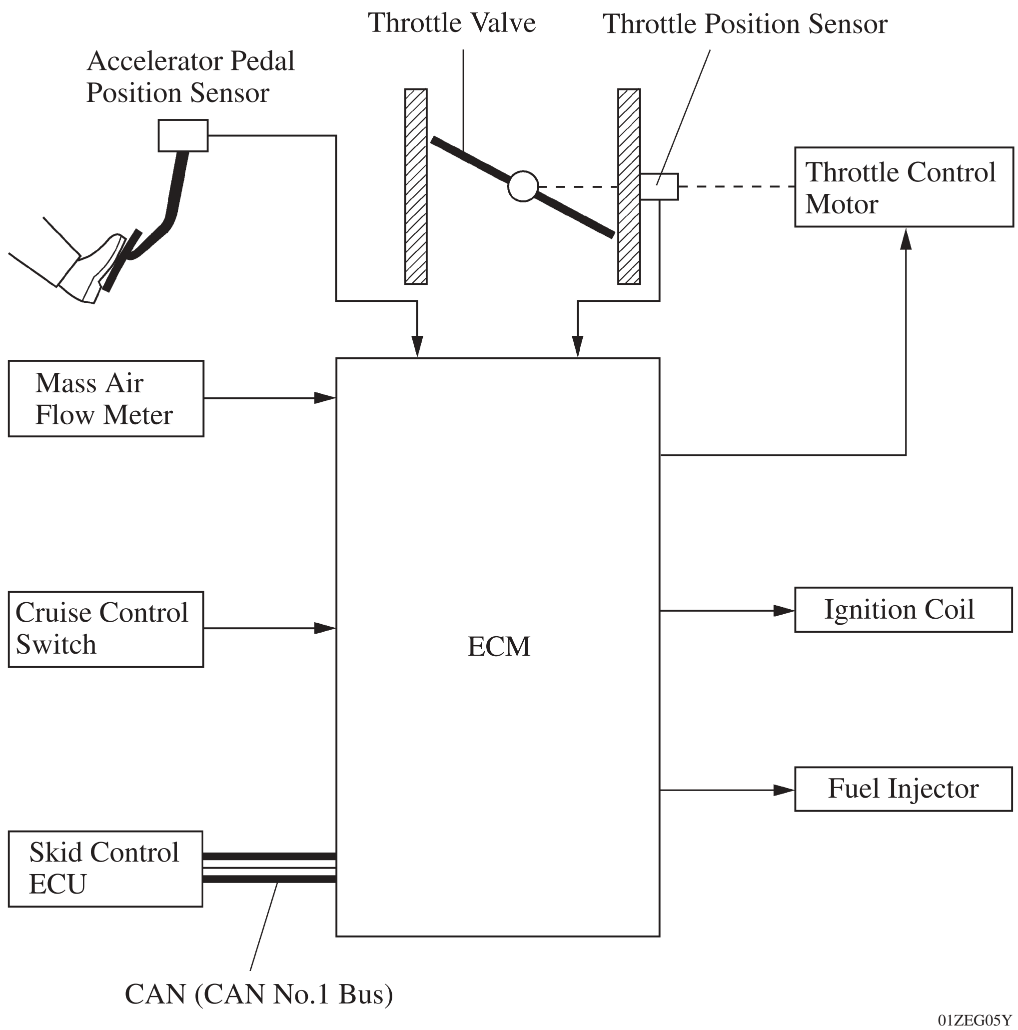
Planar Type Air Fuel Ratio Sensor **Cup Type Heated Oxygen Sensor**

6. ETCS-i (Electronic Throttle Control System-intelligent)

General

In the conventional throttle body, the throttle valve angle is determined invariably by the amount of the accelerator pedal effort. In contrast, ETCS-i uses the ECM to calculate the optimal throttle valve angle that is appropriate for the respective driving condition and uses a throttle control motor to control the angle.

▶ System Diagram ◀



Control

1) General

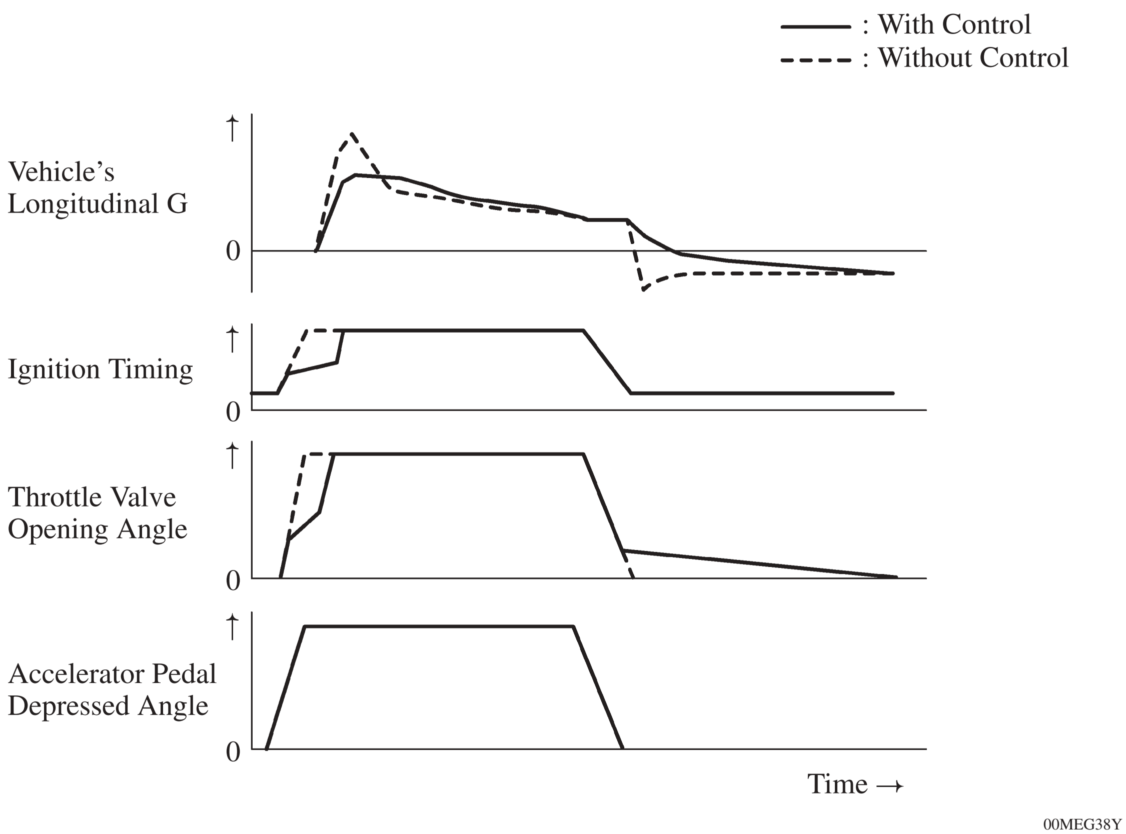
The ETCS-i consists of the following five functions:

- Normal Throttle Control (Non-linear Control)
- ISC (Idle Speed Control)
- TRAC (Traction Control)
- VSC (Vehicle Stability Control)
- Cruise Control

2) Normal Throttle Control (non-linear control)

Controls the throttle to an optimal throttle valve angle that is appropriate for the driving condition such as the amount of the accelerator pedal effort and the engine speed in order to realize excellent throttle control and comfort in all operating ranges.

▶ Conceptual Diagrams of Engine Control During Acceleration and Deceleration ◀



3) Idle Speed Control

The ECM controls the throttle valve in order to constantly maintain an ideal idle speed.

4) TRAC Throttle Control

As part of the TRAC system, the throttle valve is closed by a demand signal from the skid control ECU if an excessive amount of slippage is created at a driving wheel, thus facilitating the vehicle in ensuring excellent vehicle stability and driving force.

5) VSC Coordination Control

In order to bring the effectiveness of the VSC system control into full play, the throttle valve angle is controlled by effecting a coordination control with the skid control ECU.

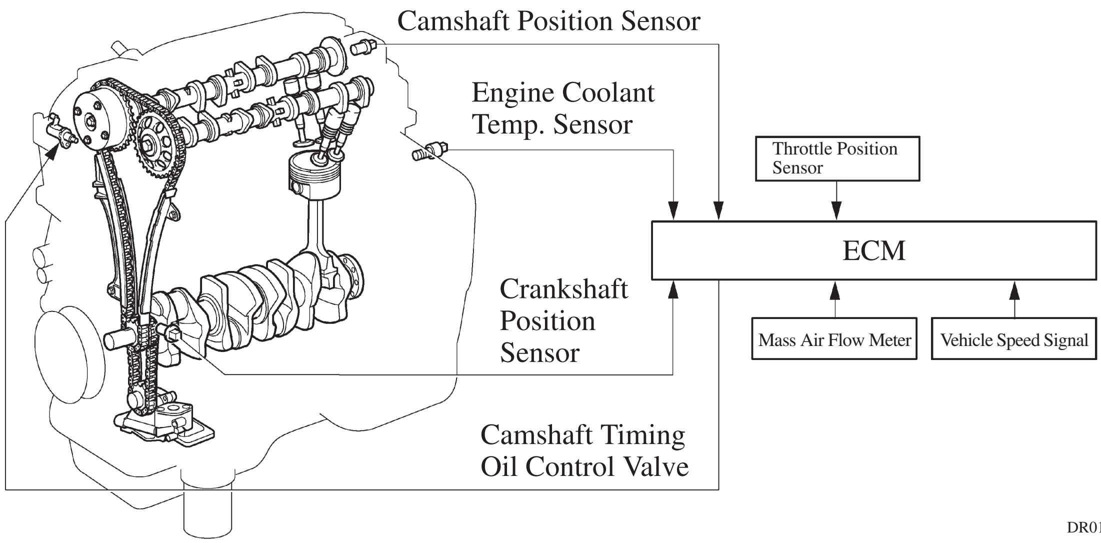
6) Cruise Control

An ECM with an integrated cruise control ECU directly actuates the throttle valve for operation of the cruise control.

7. VVT-i (Variable Valve Timing-intelligent) System

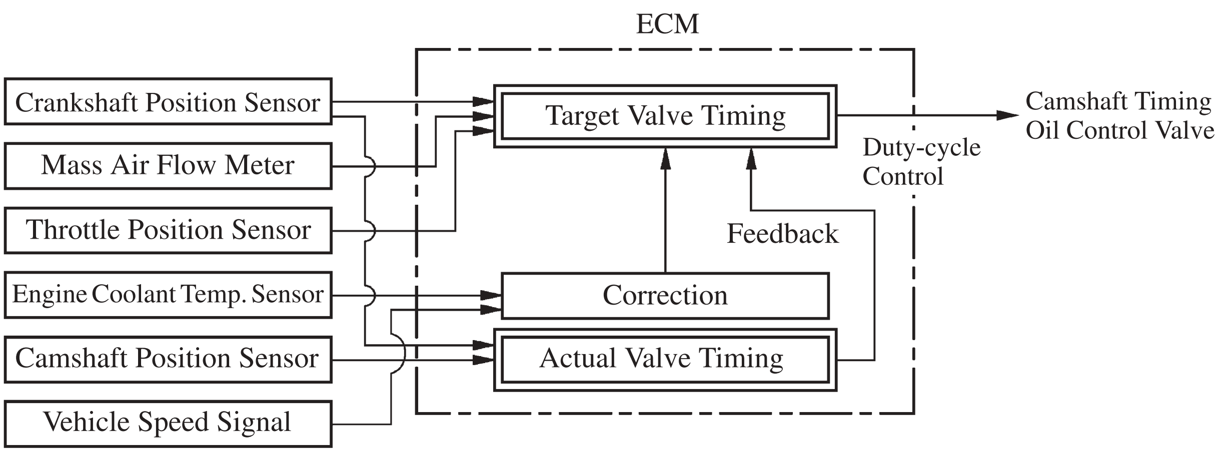
General

- The VVT-i system is designed to control the intake camshaft within a range of 40° (of Crankshaft Angle) to provide valve timing that is optimally suited to the engine condition. This improves torque in all the speed ranges as well as increasing fuel economy, and reducing exhaust emissions.



DR011EG25

- Using the engine speed, intake air volume, throttle position and water temperature, the ECM can calculate optimal valve timing for each driving condition and controls the camshaft timing oil control valve. In addition, the ECM uses signals from the camshaft position sensor and the crankshaft position sensor to detect the actual valve timing, thus providing feedback control to achieve the target valve timing.



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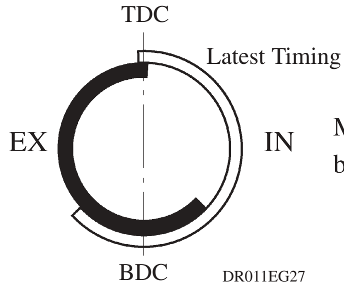
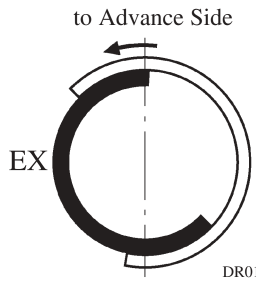
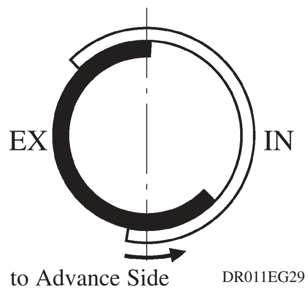
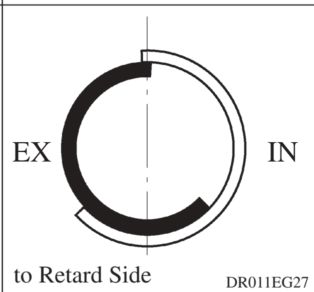
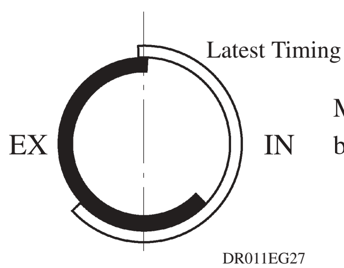
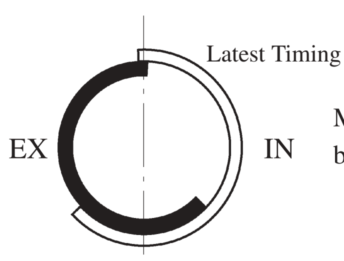
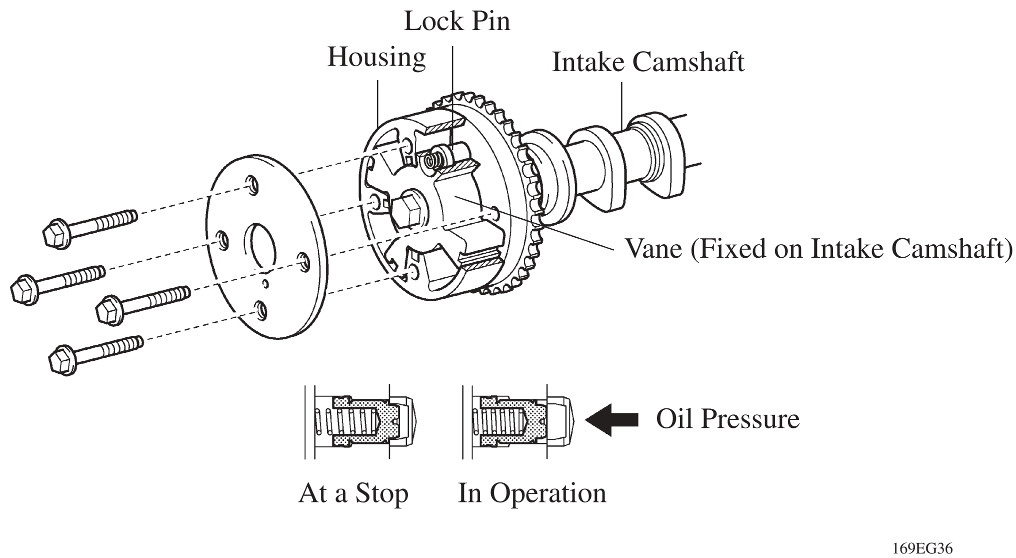
Effectiveness of the VVT-i System

Operation State	Objective		Effect
● During Idling ● At Light Load		Minimizing overlap to prevent blow back to the intake side	● Stabilized idling rpm ● Better fuel economy
At Medium Load		Increasing overlap to increase internal EGR to reduce pumping loss	● Better fuel economy ● Improved emission control
In Low to Medium Speed Range with Heavy Load		Advancing the intake valve close timing for volumetric efficiency improvement	Improved torque in low to medium speed range
In High Speed Range with Heavy Load		Retarding the intake valve close timing for volumetric efficiency improvement	Improved output
At Low Temp.		Minimizing overlap to prevent blow back to the intake side	● Stabilized fast idle rpm ● Better fuel economy
● Upon Starting ● Stopping the Engine		Minimizing overlap to prevent blow back to the intake side	Improved startability

Construction

1) VVT-i Controller

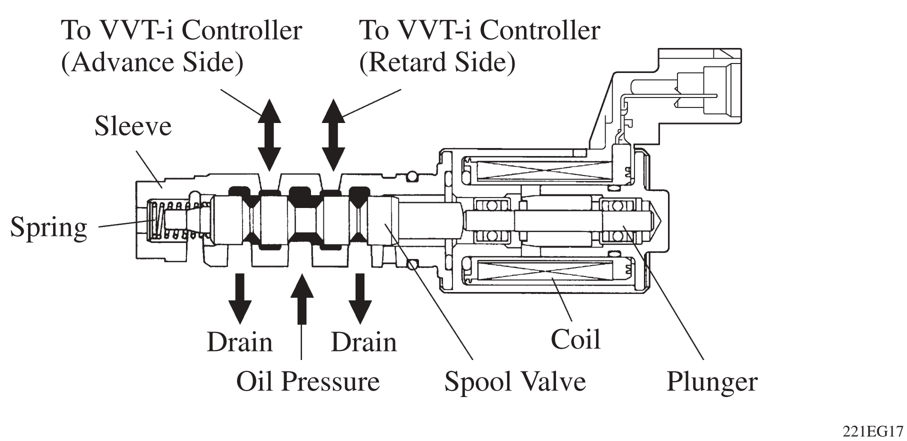
- This controller consists of the housing driven from the timing chain and the vane coupled with the intake camshaft.
- The oil pressure sent from the advance or retard side path at the intake camshaft causes rotation in the VVT-i- controller vane circumferential direction to vary the intake valve timing continuously.
  When the engine is stopped, the intake camshaft will be in the most retarded state to ensure startability.
  When hydraulic pressure is not applied to the VVT-i controller immediately after the engine has been started, the lock pin locks the movement of the VVT-i controller to prevent a knocking noise.



Lock Pin

2) Camshaft Timing Oil Control Valve

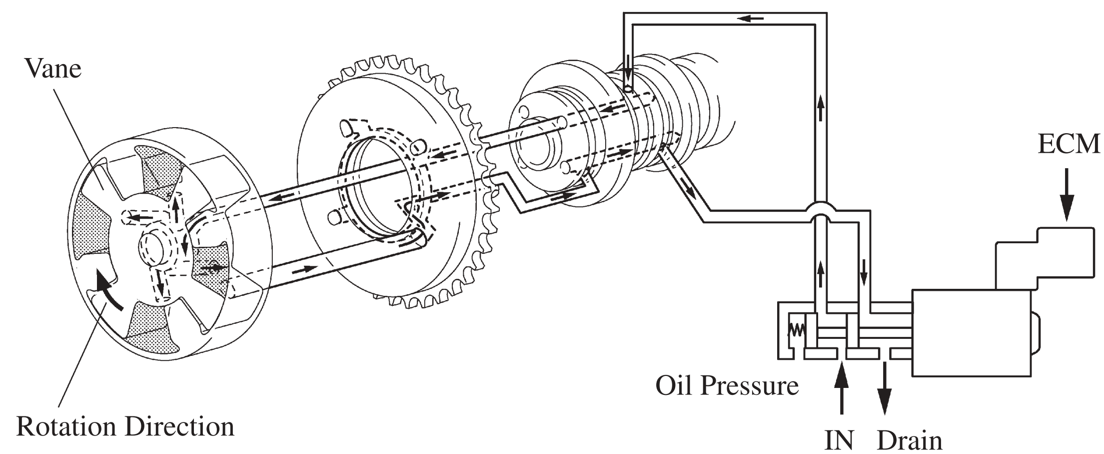
The camshaft timing oil control valve controls the spool valve position in accordance with the duty control from the ECM thus allocating the hydraulic pressure that is applied to the VVT-i controller to the advance and the retard side. When the engine is stopped, the camshaft timing oil control valve is in the most retarded state.



Operation

1) Advance

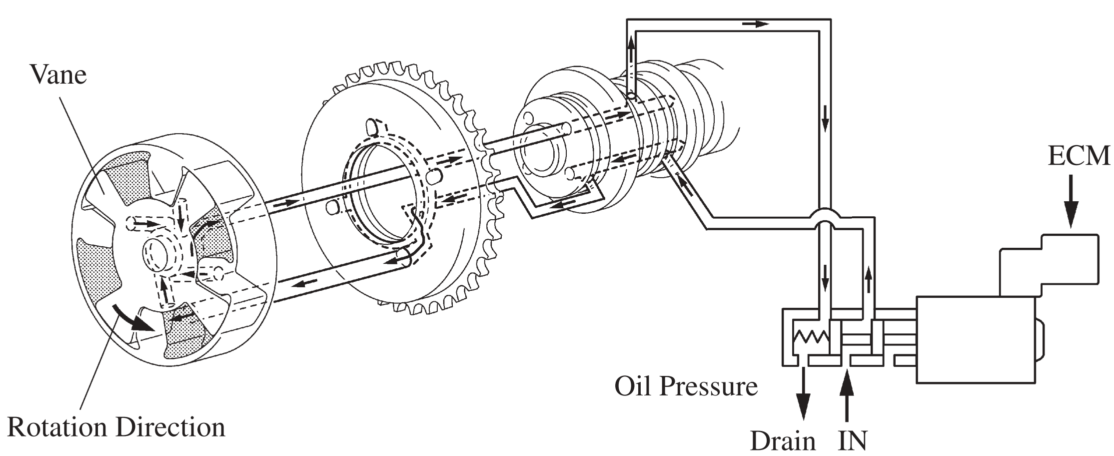
When the camshaft timing oil control valve is positioned as illustrated below by the advance signal from the ECM, the resultant oil pressure is applied to the timing advance side vane chamber to rotate the camshaft in the timing advance direction.



198EG35

2) Retard

When the camshaft timing oil control valve is positioned as illustrated below by the retard signal from the ECM, the resultant oil pressure is applied to the timing retard side vane chamber to rotate the camshaft in the timing retard direction.



198EG36

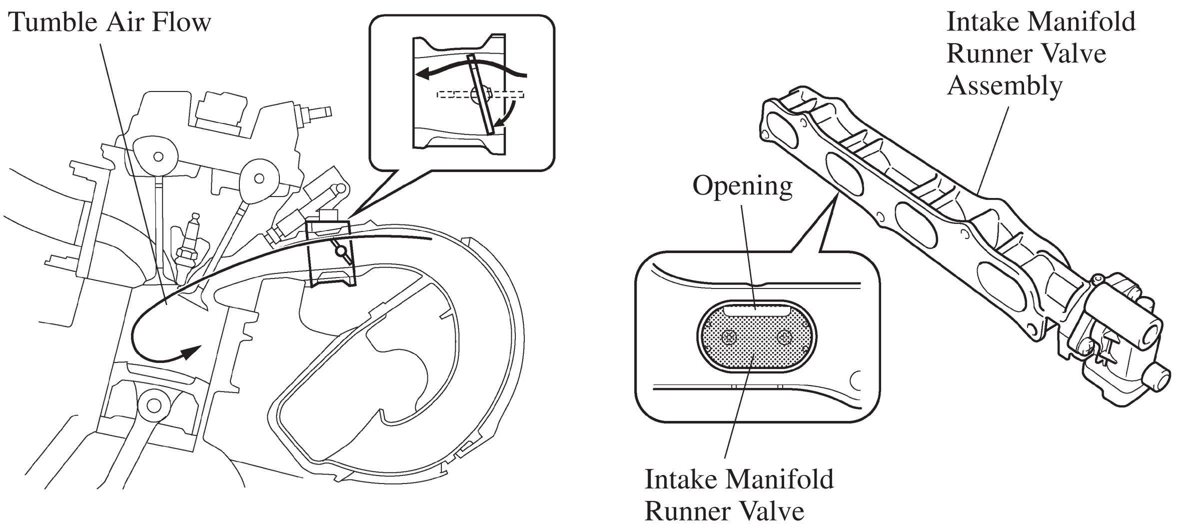
3) Hold

After reaching the target timing, the valve timing is held by keeping the camshaft timing oil control valve in the neutral position unless the traveling state changes.
This adjusts the valve timing at the desired target position and prevents the engine oil from running out when it is unnecessary.

8. Intake Manifold Runner Valve Control (California Package Model)

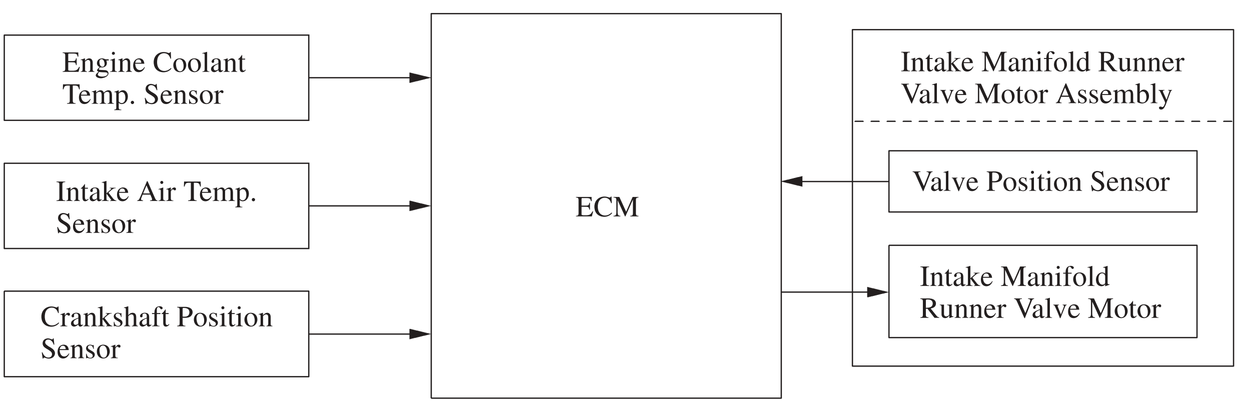
General

When the engine is cold condition and idling, this control closes the intake manifold runner valve (with an opening at the top) that is provided in the intake manifold. This creates a strong tumble airflow in the intake air that passes through the opening, which promotes the mixture of the air and fuel and improves combustion efficiency.



246EG16

▶ **System Diagram** ◀

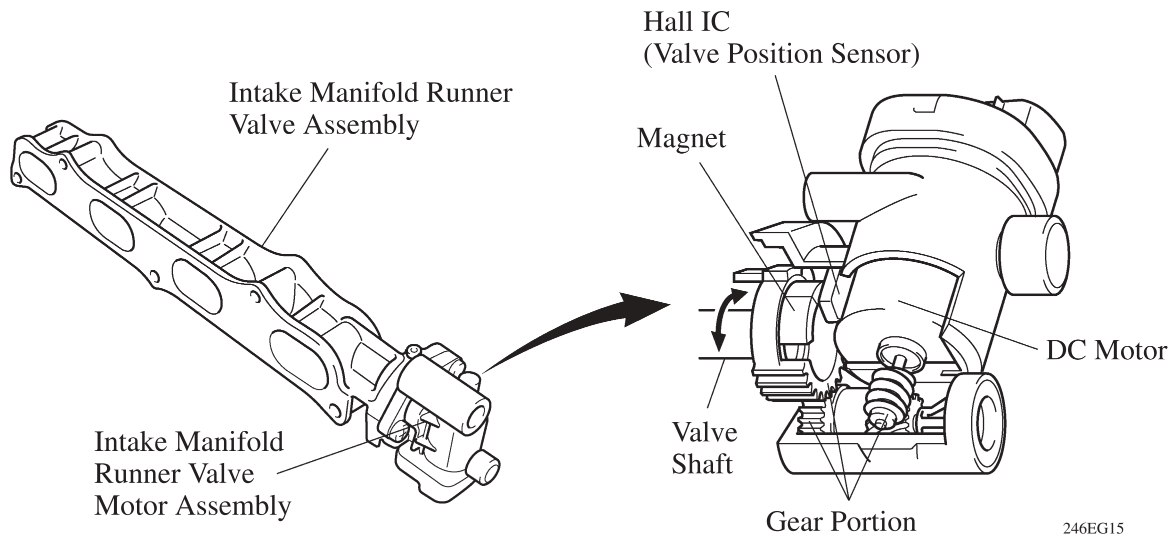


246EG39

Construction

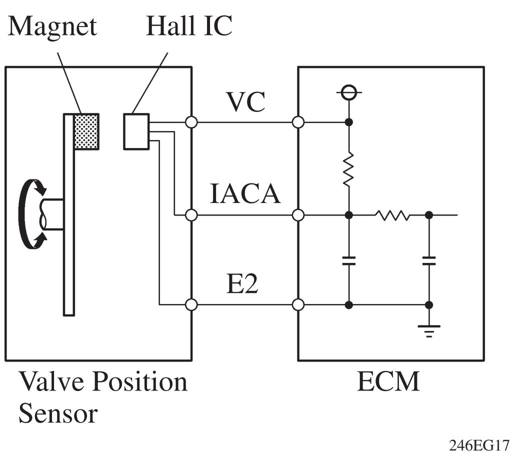
1) Intake Manifold Runner Valve Assembly

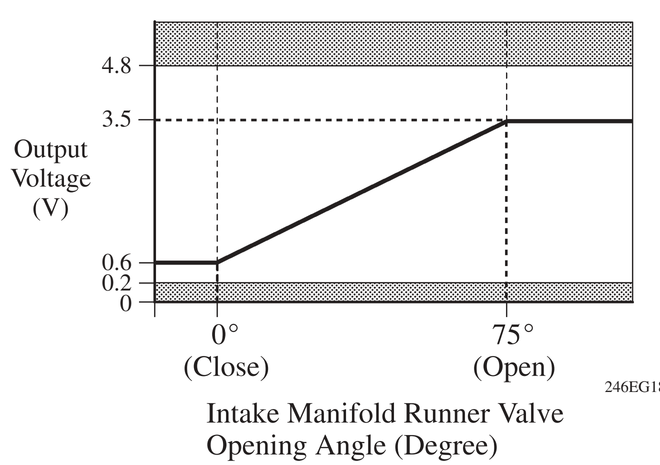
- The intake manifold runner valve assembly consists primarily of an intake manifold runner valve, valve shaft, valve body, and an intake manifold runner valve motor assembly.
- Each cylinder contains 1 intake manifold runner valve, which is mounted on the valve shaft axis.
- The intake manifold runner valve motor assembly consists primarily of a DC motor that rotates the runner valve (valve shaft), a position sensor (Hall IC) that detects the rotating position, and a gear portion that transmits the rotation of the DC motor to the valve shaft.



2) Valve Position Sensor

- The valve position sensor, which is a non-contact type that uses a Hall IC and a magnet, is mounted on the valve shaft axis.
- The valve position sensor, which is located in the intake manifold runner valve motor assembly, detects the opening position of the runner valve. This sensor gives the ECM feedback on the opening position of the runner valve, which is used for SFI control, as well as for detecting the failure of the runner valve, such as if the runner valve is stuck.
- The ECM detects a “low sensor output failure” (DTC P2016) when the output voltage of the valve position sensor is 0.2V or less, and a “high sensor output failure” (DTC P2017) when the output voltage is 4.8V or more.





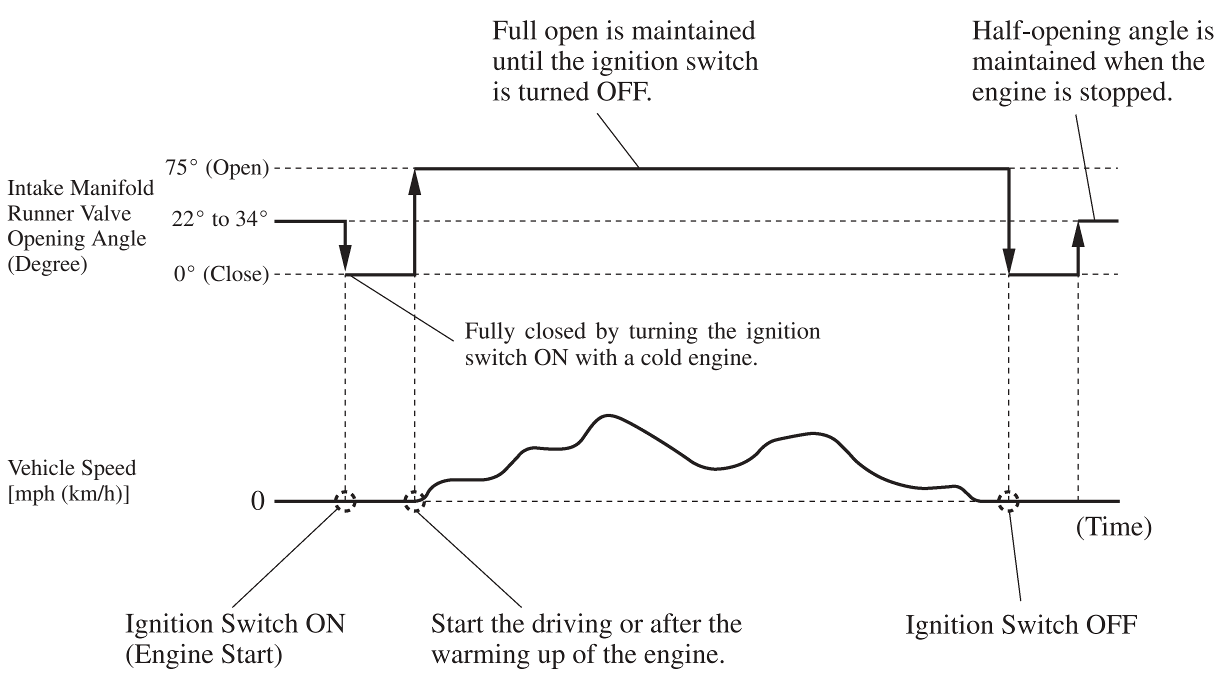
Operation

- The ECM fully closes the intake manifold runner valve during engine idling when the engine coolant temperature is between –10°C (14°F) and 60°C (140°F) and the intake air temperature is more than –10°C (14°F). Other than this condition, the ECM fully opens the intake manifold runner valve.
- When the engine is stopped, the intake manifold runner valve is kept in a half-open condition to ensure the engine startability.

▶ Open and closed condition for intake manifold runner valve ◀

Temperature of the engine coolant and the intake air when the engine starting.	Before warming up the engine.	When one of the following conditions is met: ● Throttle valve opening angle is 1.5° or more. ● Vehicle speed is 3 mph (5 km/h) or more. ● Engine speed is 3,000 rpm or more ● When the shift lever is in other than the P and N position.	After warming up of the engine
Engine coolant temp. is less than –10°C (14°F) or more than 60°C (140°F) or intake air temp. is less than –10°C (14°F).	OPEN	OPEN	OPEN
Engine coolant temp. is between –10°C (14°F) and 60°C (140°F). Intake air temp. is –10°C (14°F) or more.	CLOSED	OPEN	OPEN

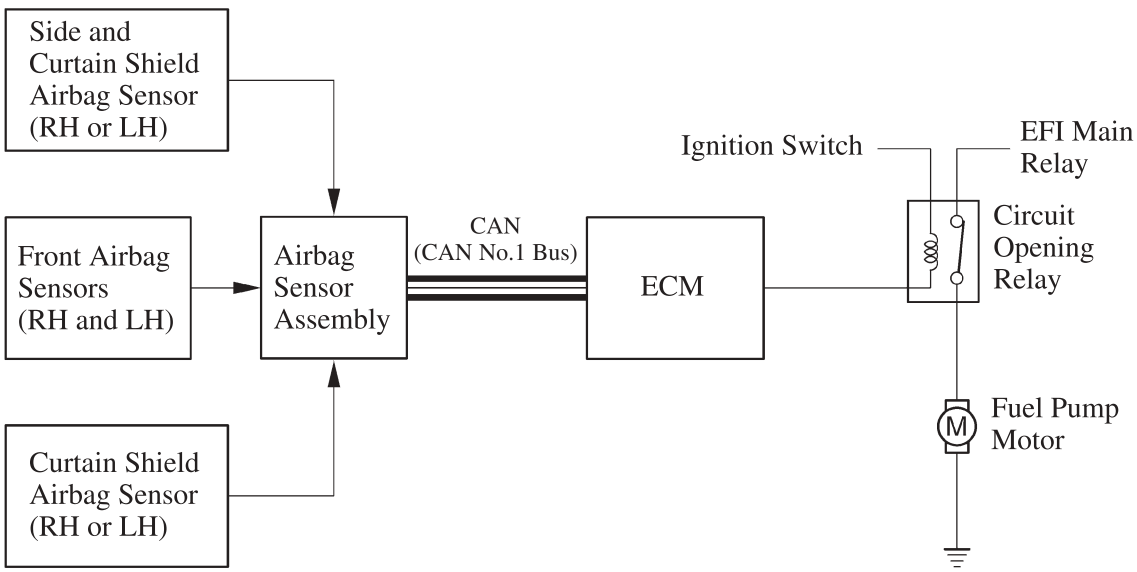
▶ Intake manifold runner valve operation (Normal condition) ◀



246EG19

9. Fuel Pump Control

A fuel cut control is used to stop the fuel pump once when any of the SRS airbags is deployed. In this system, the airbag deployment signal from the airbag sensor assembly is detected by the ECM, and it turns OFF the circuit opening relay. After the fuel cut control has been activated, turning the ignition switch from OFF to ON cancels the fuel cut control, and the engine can be restarted.



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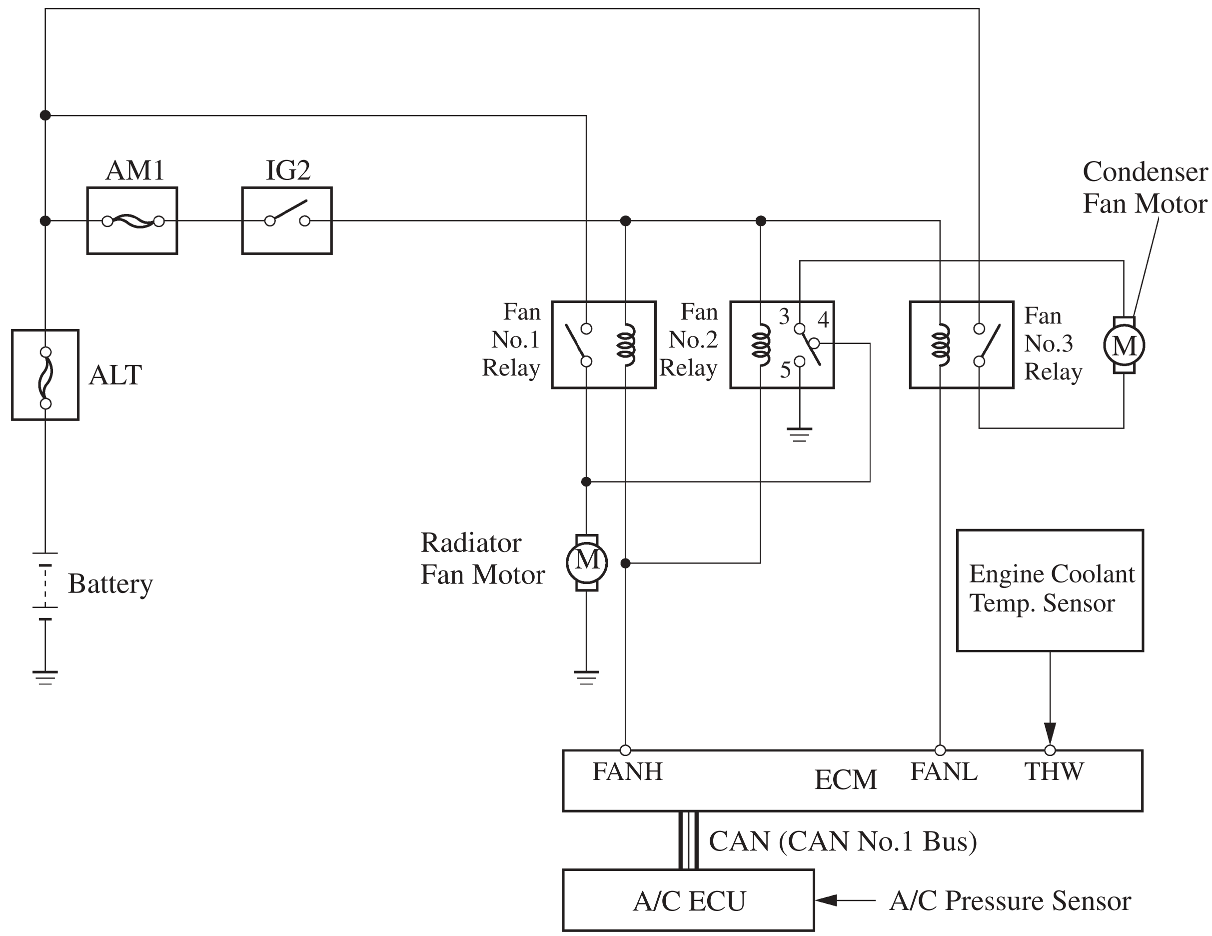
10. Cooling Fan Control System

A cooling fan control system in which the ECM controls the cooling fan speed in accordance with the engine coolant temperature and the air conditioning operating condition.

- The ECM controls the cooling fan speed based on A/C pressure sensor signals and engine coolant temperature sensor signals. The A/C pressure sensor signals are sent from the A/C ECU to the ECM via the CAN. This control is accomplished by operating the 2 fan motors in 2 stages at low speed (series connection) and high speed (parallel connection).

▶ Wiring Diagram ◀



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▶ Cooling Fan Operation ◀

Air Conditioning Operating Condition	Engine Coolant Temperature	Relay Operation			Cooling Fan Motor Connection	Cooling Fan Operation
		No.1	No.2	No.3		
OFF	Low	OFF	3 to 4	OFF	OFF	OFF
	High	ON	3 to 5	ON	Parallel	High
A/C Pressure "Low"	Low	OFF	3 to 4	ON	Series	Low
A/C Pressure "High"	Low	ON	3 to 5	ON	Parallel	High
A/C Pressure "Low"	High	ON	3 to 5	ON	Parallel	High
A/C Pressure "High"	High	ON	3 to 5	ON	Parallel	High

11. Evaporative Emission Control System

General

The evaporative emission control system prevents the fuel vapors that are created in the fuel tank from being released directly into the atmosphere.
The canister stores the fuel vapors that have been created in the fuel tank.

- The ECM controls the purge VSV in accordance with the driving conditions in order to direct the fuel vapors into the engine, where they are burned.
- In this system, the ECM checks for evaporative emission leaks and outputs DTC (Diagnostic Trouble Code) in the event of a malfunction. An evaporative emission leak check consists of an application of vacuum to the evaporative emissions system and monitoring the system for changes in pressure in order to detect a leakage.
- This system consists of a purg VSV, canister, refueling valve, canister pump module, and ECM.
- An ORVR (Onboard Refueling Vapor Recovery) function is provided in the refueling valve.
- The canister pressure sensor has been included to the canister pump module.
- An air filter has been provided on the fresh air line. This air filter is maintenance-free.
- The EVAP service port has been removed.
- The following are the typical conditions necessary to enable an evaporative emission leak check:

Typical Enabling Condition	● Five hours have elapsed after the engine has been turned OFF*. ● Altitude: Below 2400 m (8000 feet) ● Battery Voltage: 10.5 V or more ● Ignition switch: OFF ● Engine Coolant Temperature: 4.4 to 35°C (40 to 95°F) ● Intake Air Temperature: 4.4 to 35°C (40 to 95°F)

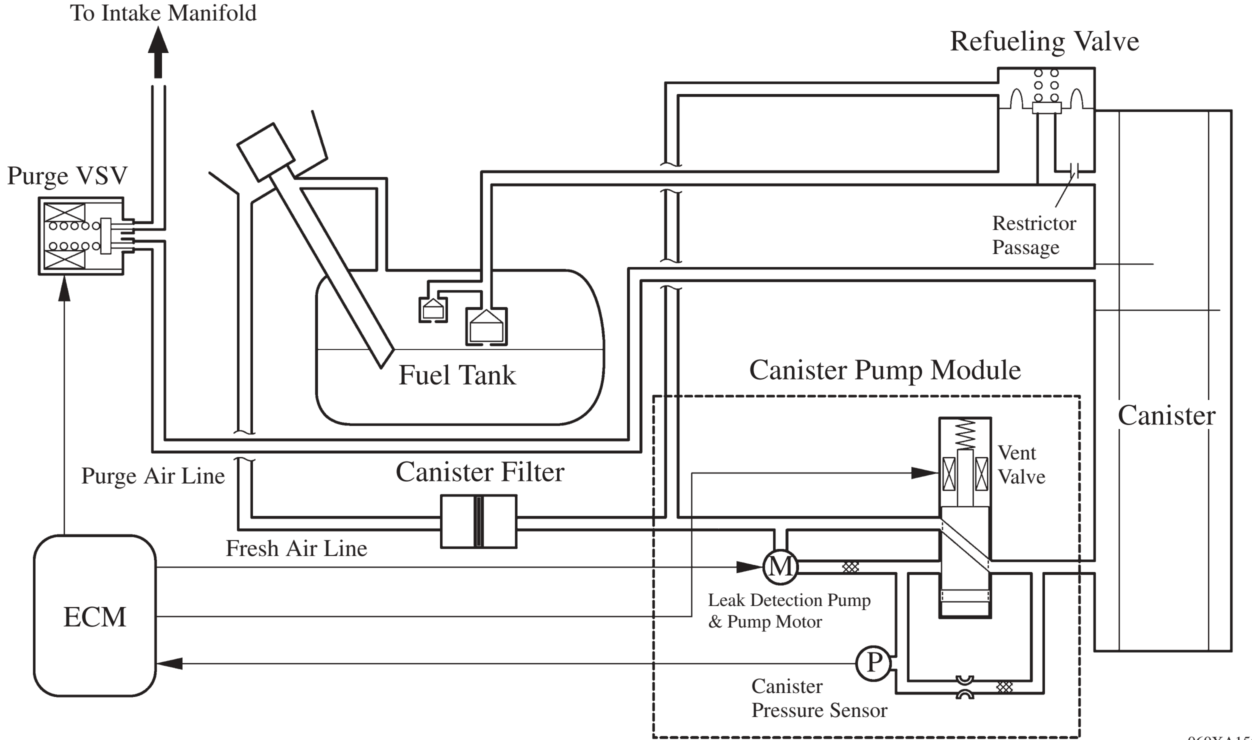
*: If engine coolant temperature does not drop below 35°C (95°F), this time should be extended to 7 hours. Even after that, if the temperature is not less than 35°C (95°F), the time should be extended to 9.5 hours.

Service Tip

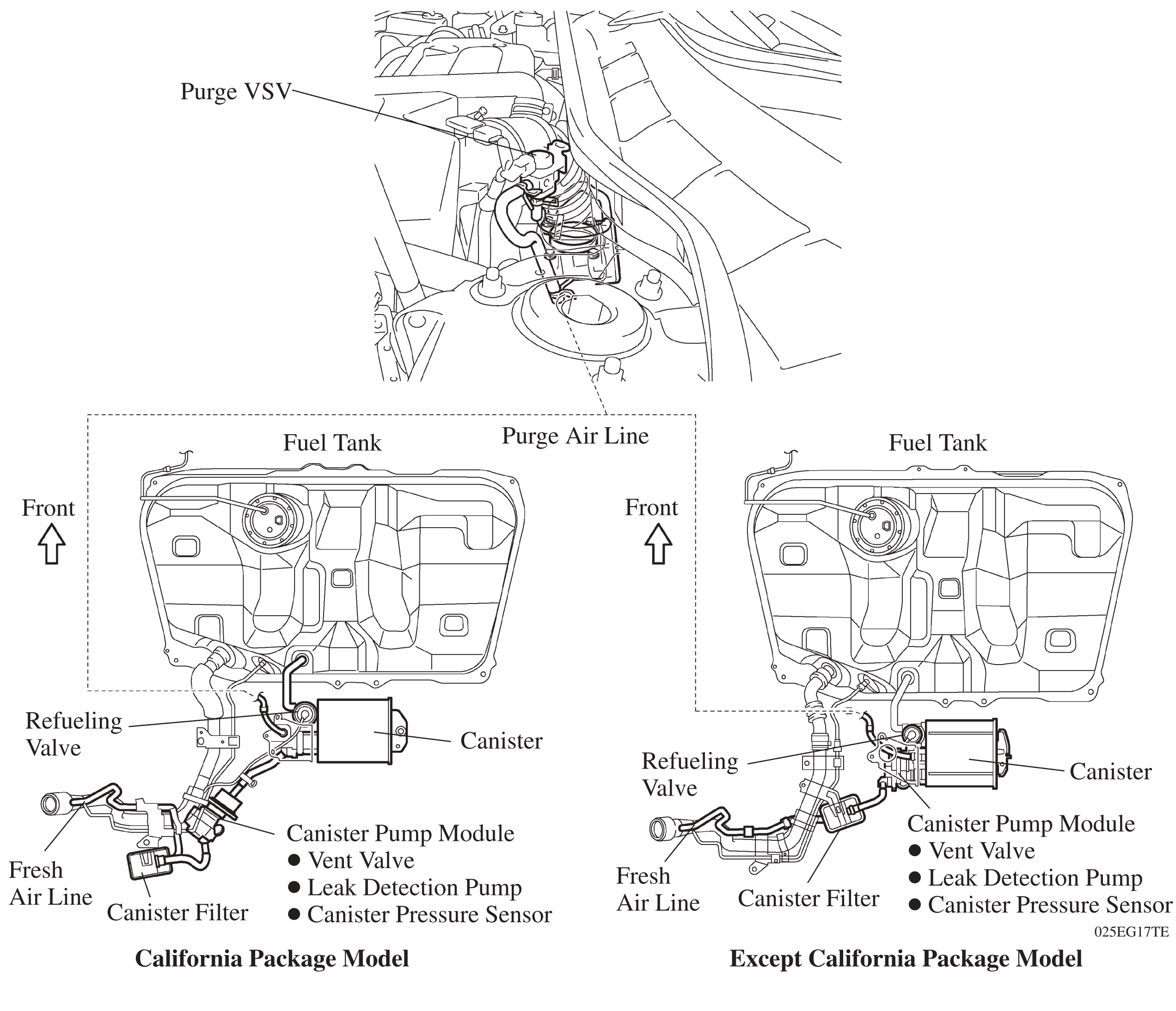
The canister pump module performs a fuel evaporative emission leakage check. This check is done approximately five hours after the engine is turned off. Sound may be heard coming from underneath the luggage compartment for several minutes. This does not indicate a malfunction.

- Pinpoint pressure test procedure is adopted by pressurizing the fresh air line that runs from the canister pump module to the air filler neck. For details, see the 2007 Camry Repair Manual (Pub. No. RM0250U).

System Diagram



Layout of Main Components



California Package Model **Except California Package Model**

Function of Main Components

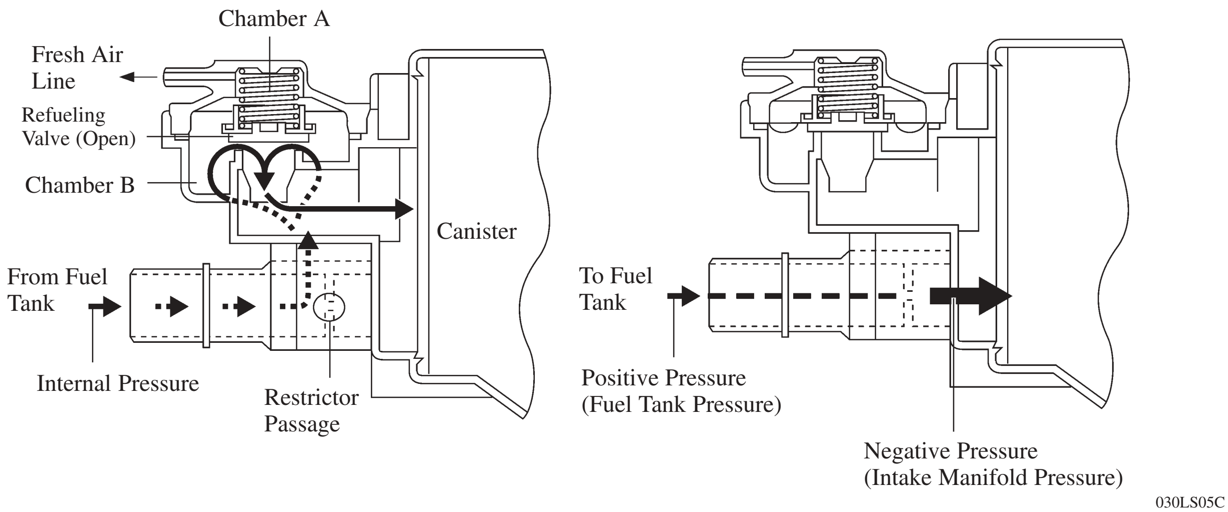
<table>
<tr><th colspan="2">Component</th><th>Function</th></tr>
<tr><td colspan="2">Canister</td><td>Contains activated charcoal to absorb the fuel vapors that are created in the fuel tank.</td></tr>
<tr><td rowspan="2">Refueling Valve</td><td></td><td>Controls the flow rate of the fuel vapors from the fuel tank to the canister when the system is purging or during refueling.</td></tr>
<tr><td>Restrictor Passage</td><td>Prevents a large amount of vacuum during purge operation or system monitoring operation from affecting the pressure in the fuel tank.</td></tr>
<tr><td colspan="2">Fresh Air Line</td><td>Fresh air goes into the canister and the cleaned drain air goes out into the atmosphere.</td></tr>
<tr><td rowspan="3">Canister Pump Module</td><td>Canister Vent Valve</td><td>Opens and closes the fresh air line in accordance with the signals from the ECM.</td></tr>
<tr><td>Leak Detection Pump & Pump Motor</td><td>Applies vacuum pressure to the evaporative emission system in accordance with the signals from the ECM.</td></tr>
<tr><td>Canister Pressure Sensor</td><td>Detects the pressure in the evaporative emission system and sends the signals to the ECM.</td></tr>
<tr><td colspan="2">Purge VSV</td><td>Opens in accordance with the signals from the ECM when the system is purging, in order to send the fuel vapors that were absorbed by the canister into the intake manifold. In system monitoring mode, this valve controls the introduction of the vacuum into the fuel tank.</td></tr>
<tr><td colspan="2">Canister Filter</td><td>Prevents dust and debris in the fresh air from entering the system.</td></tr>
<tr><td colspan="2">ECM</td><td>Controls the canister pump module and the purge VSV in accordance with the signals from various sensors, in order to achieve a purge volume that suits the driving conditions. In addition, the ECM monitors the system for any leakage and outputs a DTC if a malfunction is found.</td></tr>
</table>

Construction and Operation

1) Refueling Valve

The refueling valve consists of chamber A, chamber B, and the restrictor passage. A constant atmospheric pressure is applied to chamber A.

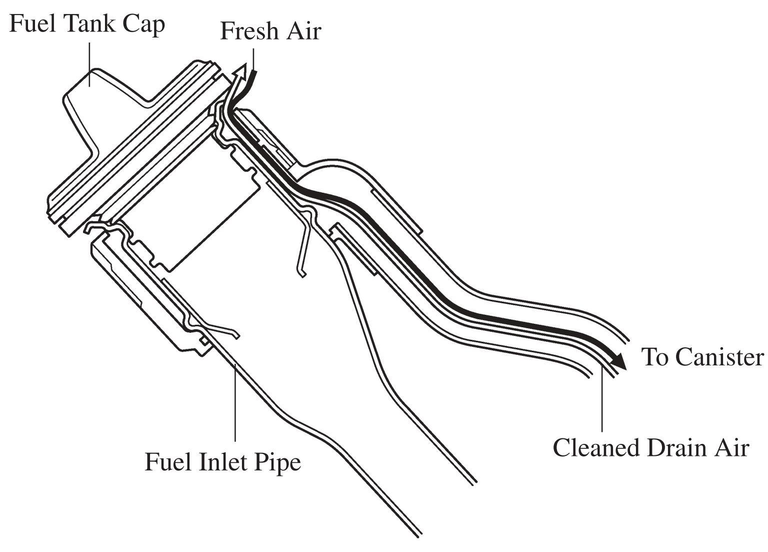
- During refueling, the internal pressure of the fuel tank increases. This pressure causes the refueling valve to lift up, allowing the fuel vapors to enter the canister.
- The restrictor passage prevents the large amount of vacuum that is created during purge operation or system monitoring operation from entering the fuel tank, and limits the flow of the fuel vapors from the fuel tank to the canister. If a large volume of fuel vapors enters the intake manifold, it will affect the air-fuel ratio control of the engine. Therefore, the role of the restrictor passage is to help prevent this from occurring.



During Refueling **During Purge Operation or System Monitoring Operation**

2) Fuel Inlet (Fresh Air Inlet)

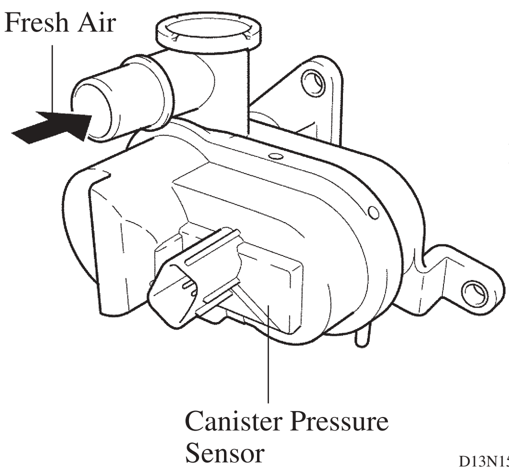
In accordance with the change of structure of the evaporative emission control system, the location of the fresh air line inlet has been changed from the air cleaner to the near the fuel inlet. The fresh air from the atmosphere and drain air cleaned by the canister will go in or out of the system through the passages shown below.



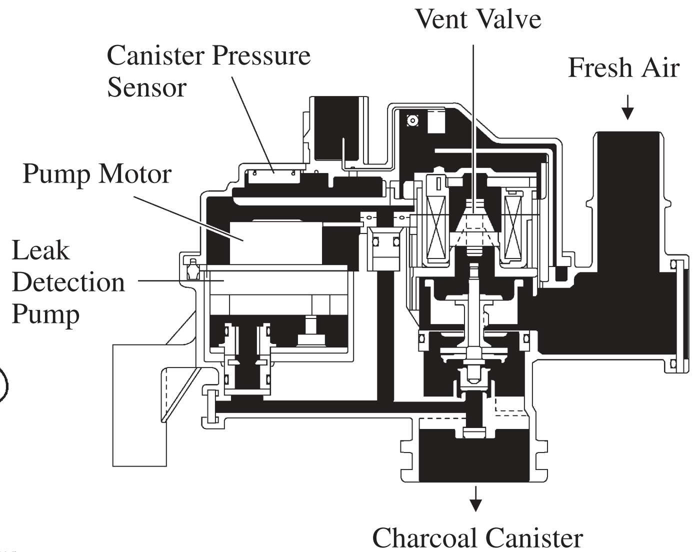
3) Canister Pump module

The canister Pump module consists of the vent valve, canister pressure sensor, leak detection pump and pump motor.

- The vent valve switches the passages in accordance with the signals received from the ECM.
- A DC type brushless motor is used for the pump motor.
- A vane type leak detection pump is used.

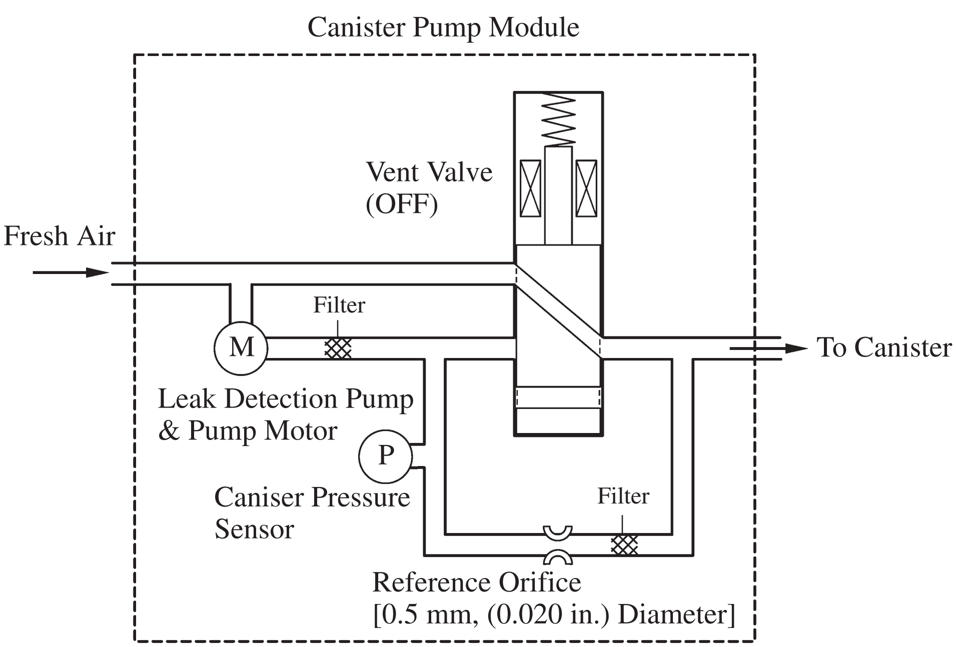


D13N15



D13N16

▶ Simple Diagram ◀



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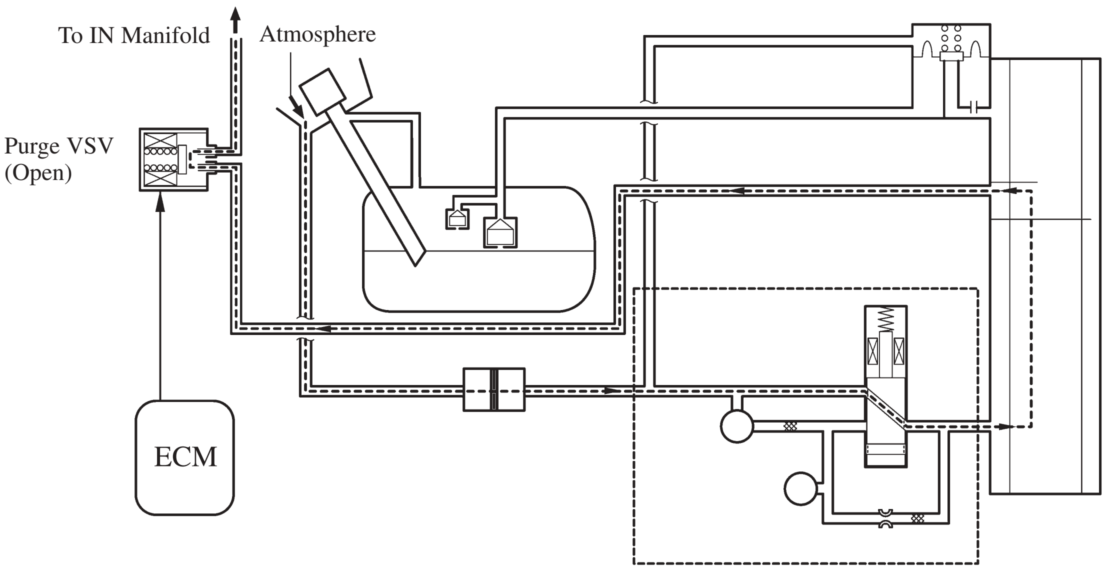
System Operation

1) Purge Flow Control

When the engine has reached predetermined parameters (closed loop, engine coolant temp. above 80°C (176°F), etc), stored fuel vapors are purged from the canister whenever the purge VSV is opened by the ECM.
The ECM will change the duty ratio cycle of the purge VSV, thus controlling purge flow volume. Purge flow volume is determined by intake manifold pressure and the duty ratio cycle of the purge VSV. Atmospheric pressure is allowed into the canister to ensure that purge flow is constantly maintained whenever purge vacuum is applied to the canister.



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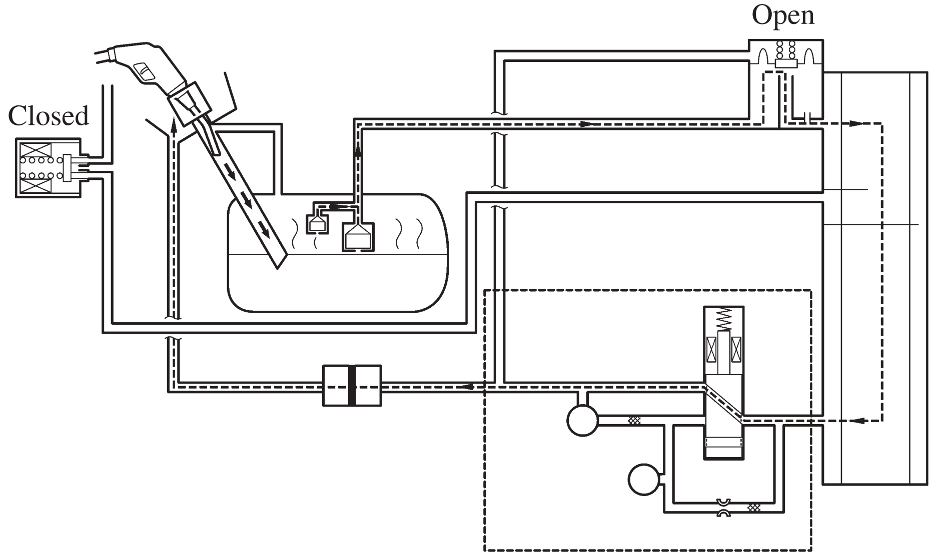
2) ORVR (On-Board Refueling Vapor Recovery)

When the internal pressure of the fuel tank increases during refueling, this pressure causes the diaphragm in the refueling valve to lift up, allowing the fuel vapors to enter the canister. The air that has had the fuel vapors removed from it will be discharged through the fresh air line. The vent valve is used to open and close the fresh air line, and it is always open (even when the engine is stopped) except when the vehicle is in monitoring mode (the valve will be open as long as the vehicle is not in monitoring mode). If the vehicle is refueled in system monitoring mode, the ECM will recognize the refueling by way of the canister pressure sensor, which detects the sudden pressure increase in the fuel tank, and the ECM will open the vent valve.



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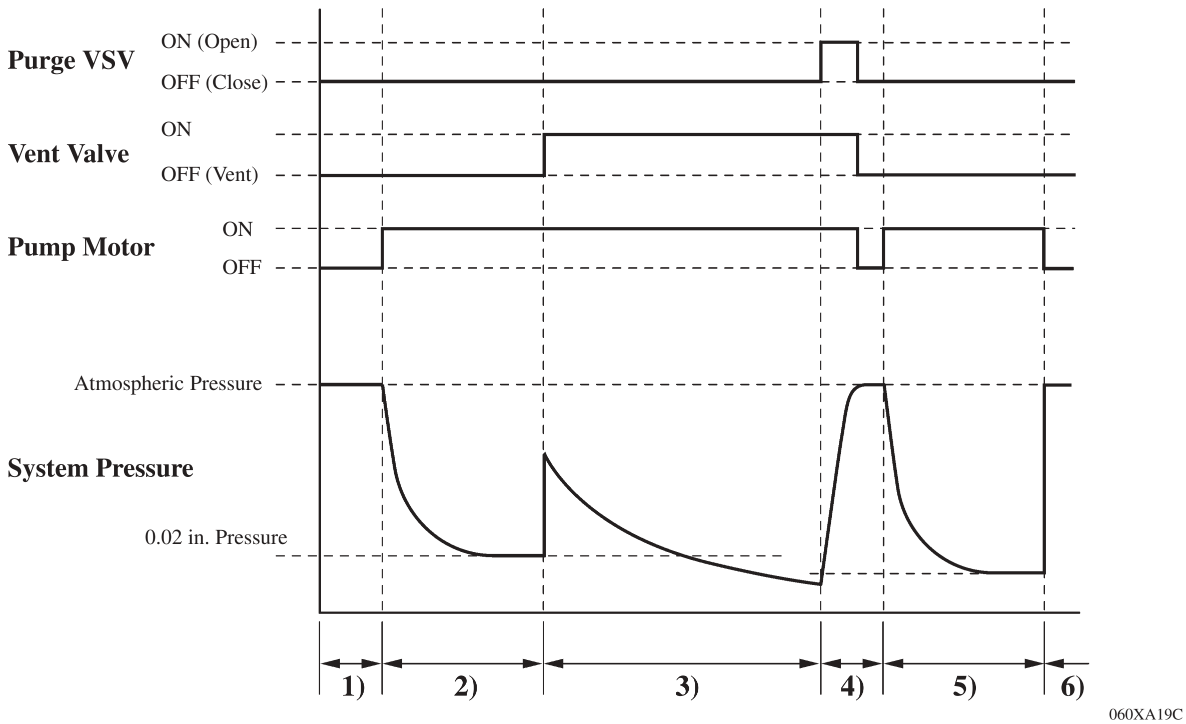
3) EVAP Leak Check

a. General

The EVAP leak check operates in accordance with the following timing chart:

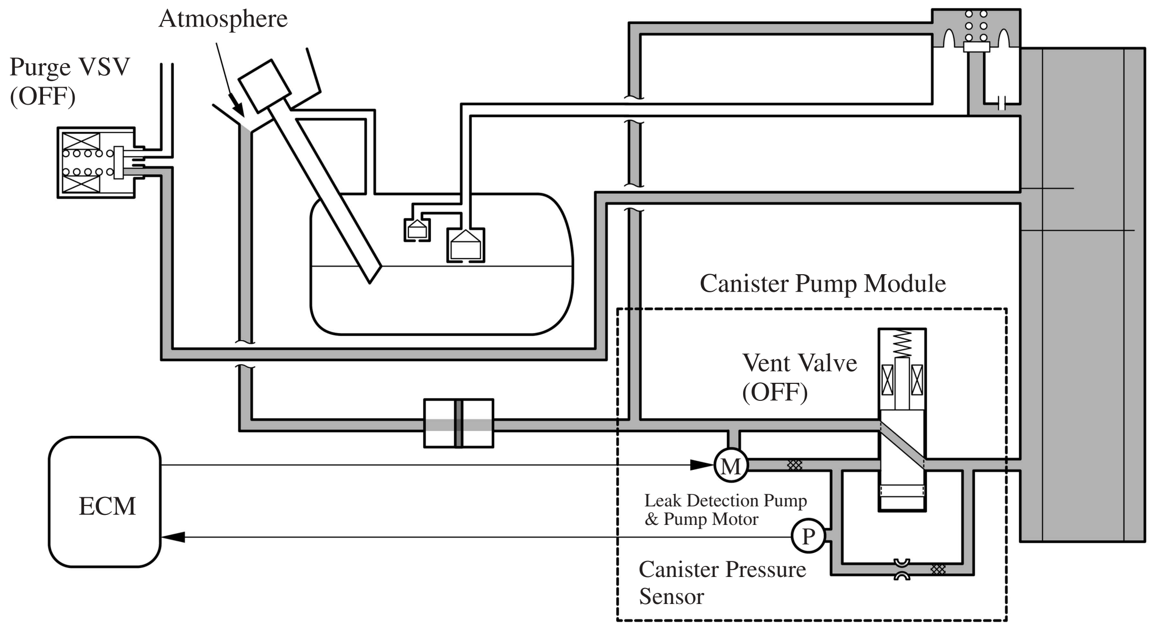
▶ Timing Chart ◀



Order	Operation	Description	Time
1)	Atmospheric Pressure Measurement	The ECM turns the vent valve OFF (vent) and measures EVAP system pressure to memorize the atmospheric pressure.	—
2)	0.02 in. Leak Pressure Measurement	The leak detection pump creates negative pressure (vacuum) through a 0.02 in. orifice and the pressure is measured. The ECM determines this as the 0.02 in. leak pressure.	20 sec.
3)	EVAP Leak Check	The leak detection pump creates negative pressure (vacuum) in the EVAP system and the EVAP system pressure is measured. If the stabilized pressure is larger than the 0.02 in. leak pressure, ECM determines that the EVAP system has a leak. If the EVAP pressure does not stabilize within 15 minutes, the ECM cancels EVAP monitor.	Within 15 min.
4)	Purge VSV Monitor	The ECM opens the purge VSV and measures the EVAP pressure increase. If the increase is large, the ECM interprets this as normal.	10 sec.
5)	Repeat 0.02 in. Leak Pressure Measurement	The leak detection pump creates negative pressure (vacuum) through the 0.02 in. orifice and the pressure is measured. The ECM determines this as the 0.02 in. leak pressure.	20 sec.
6)	Final Check	The ECM measures the atmospheric pressure and records the monitor result.	—

b. Atmospheric Pressure Measurement

1) When the ignition switch is turned OFF, the purge VSV and the vent valve are turned OFF. Therefore, atmospheric pressure is introduced into the canister.
2) The ECM measures the atmospheric pressure based on the signals provided by the canister pressure sensor.
3) If the measurement value is out of standards, the ECM actuates the leak detection pump in order to monitor the changes in the pressure.



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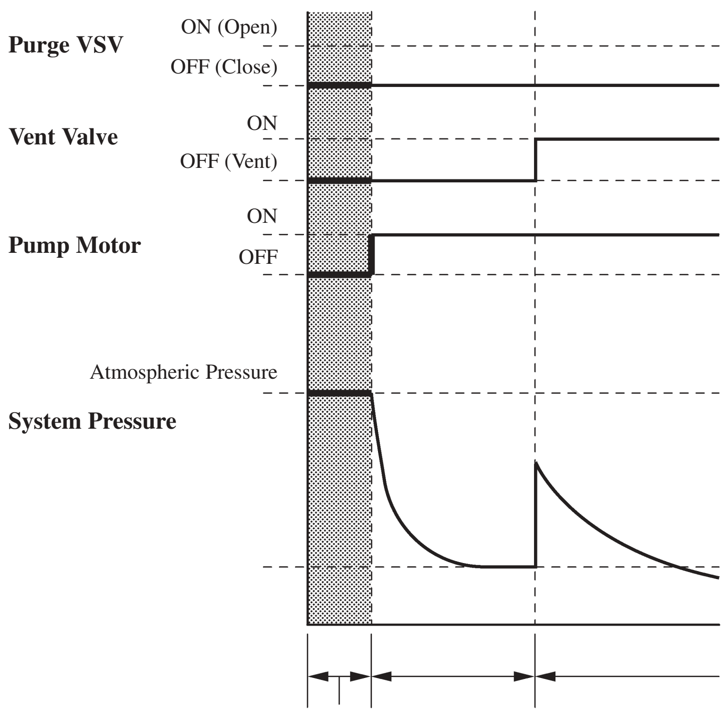
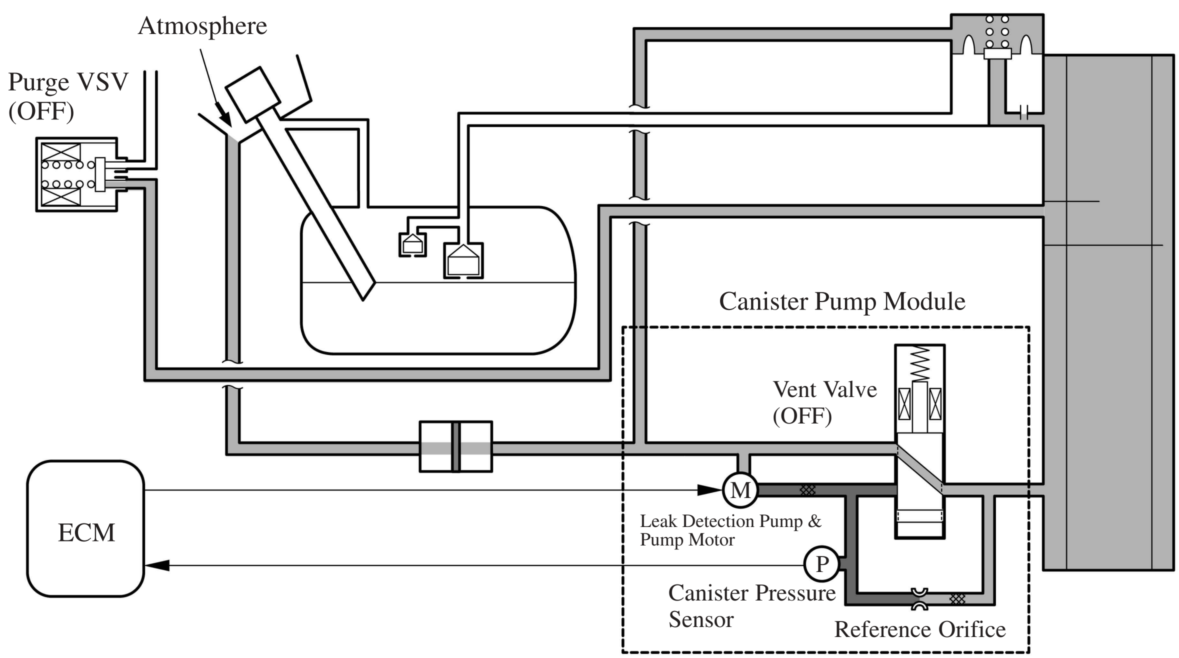


Atmospheric Pressure Measurement

D13N22

c. 0.02 in. Leak Pressure Measurement

1) The vent valve remains off, and the ECM introduces atmospheric pressure into the canister and actuates the leak detection pump in order to create a negative pressure.
2) At this time, the pressure will not decrease beyond a 0.02 in. pressure due to the atmospheric pressure that enters through a 0.02 in. diameter reference orifice.
3) The ECM compares the logic value and this pressure, and stores it as a 0.02 in. leak pressure in its memory.
4) If the measurement value is below the standard, the ECM will determine that the reference orifice is clogged and store DTC (Diagnostic Trouble Code) P043E in its memory.
5) If the measurement value is above the standard, the ECM will determine that a high flow rate pressure is passing through the reference orifice and store DTC (Diagnostic Trouble Code) P043F, P2401 and P2402 in its memory.



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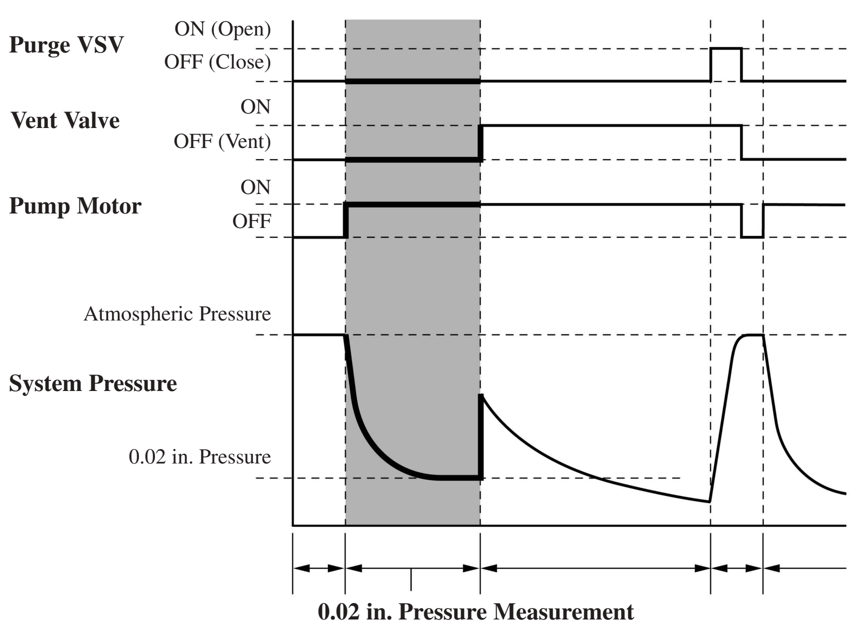

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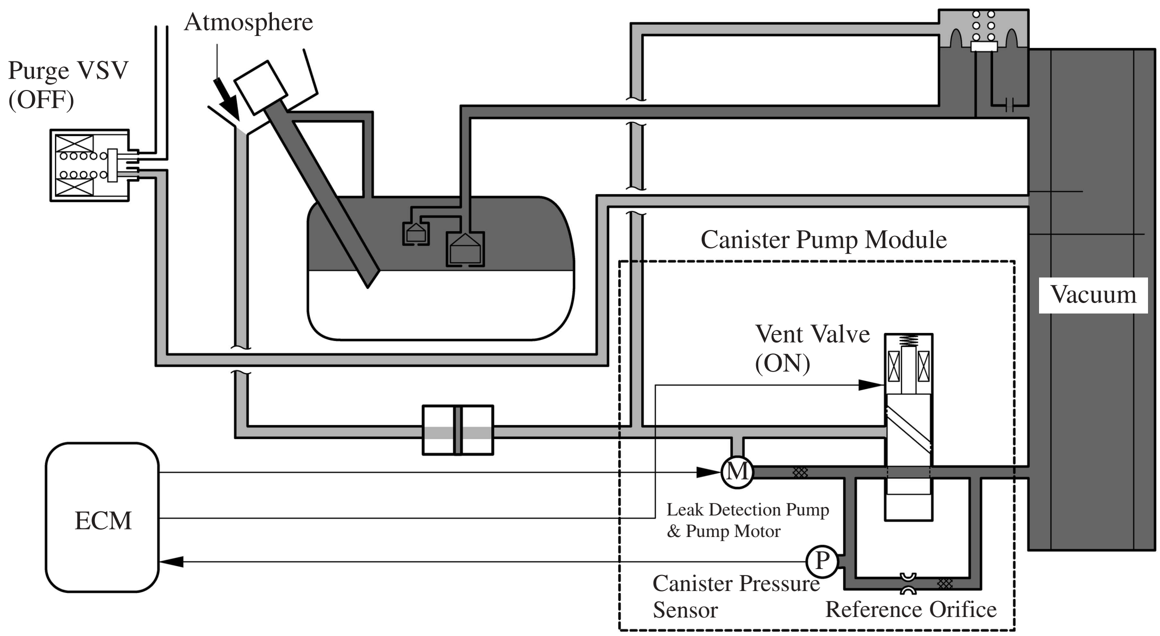
d. EVAP Leak Check

1) While actuating the leak detection pump, the ECM turns ON the vent valve in order to introduce a vacuum into the canister.
2) When the pressure in the system stabilizes, the ECM compares this pressure and the 0.02 in. pressure in order to check for a leakage.
3) If the detection value is below the 0.02 in. pressure, the ECM determines that there is no leakage.
4) If the detection value is above the 0.02 in. pressure and near atmospheric pressure, the ECM determines that there is a gross leakage (large hole) and stores DTC P0455 in its memory.
5) If the detection value is above the 0.02 in. pressure, the ECM determines that there is a small leakage and stores DTC P0456 in its memory.



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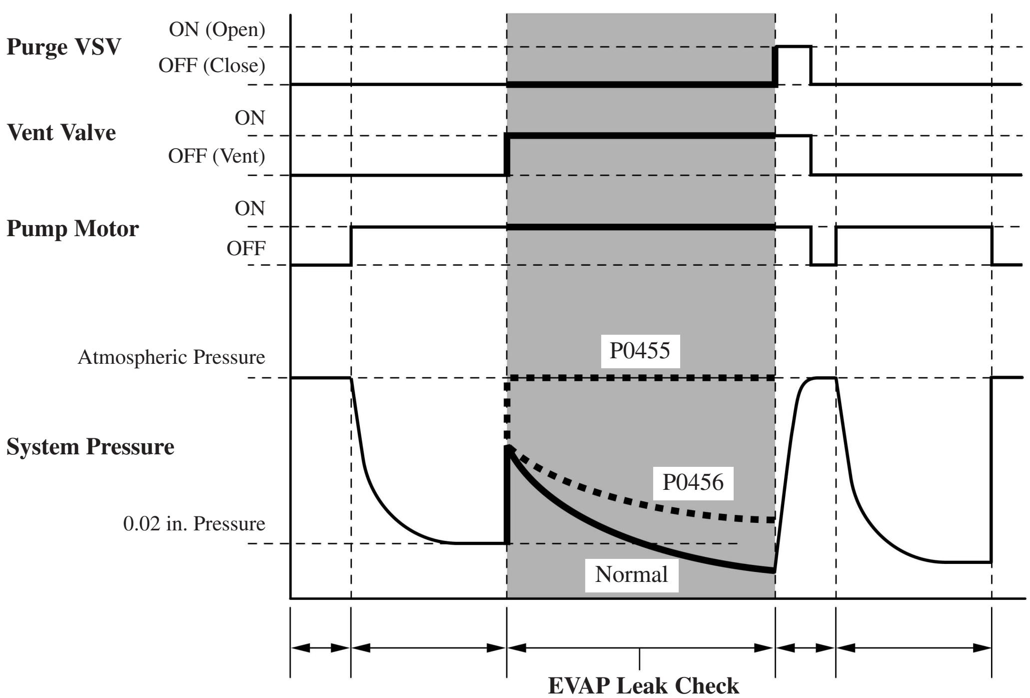

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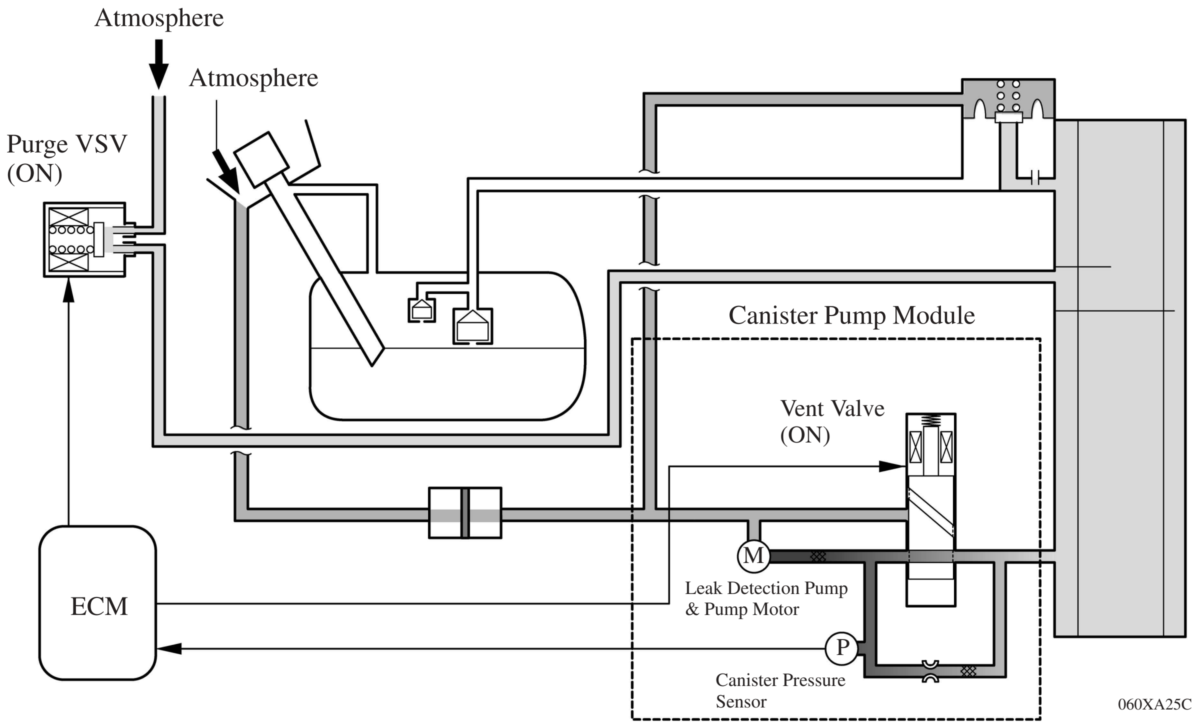
e. Purge VSV Monitor

1) After completing an EVAP leak check, the ECM turns ON (open) the purge VSV with the leak detection pump actuated, and introduces the atmospheric pressure from the intake manifold to the canister.
2) If the pressure change at this time is within the normal range, the ECM determines the condition to be normal.
3) If the pressure is out of the normal range, the ECM will stop the purge VSV monitor and store DTC P0441 in its memory.



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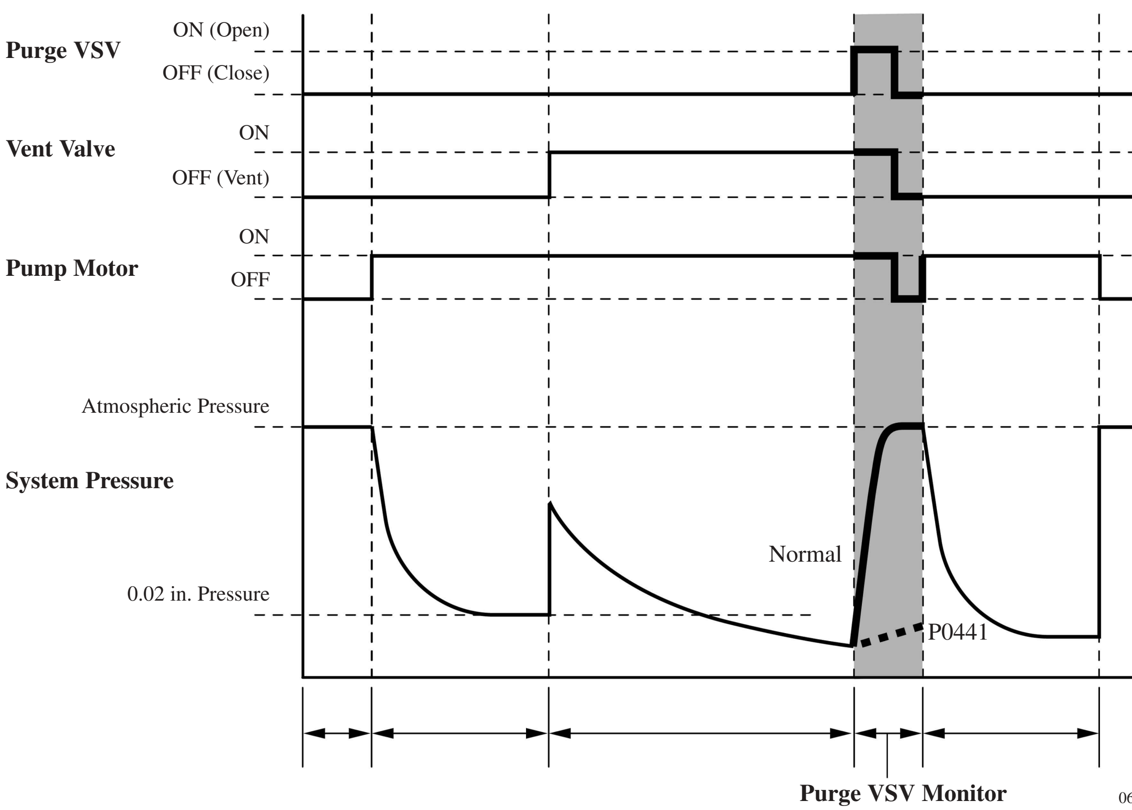
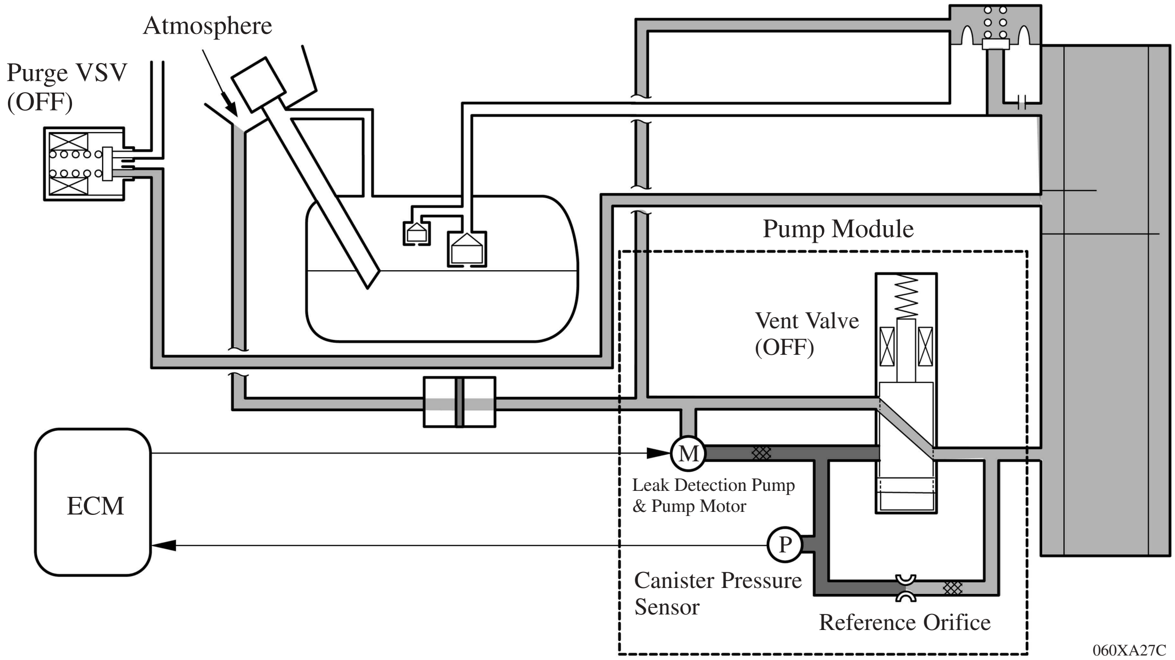


060XA26C

f. Repeat 0.02 in. Leak Pressure Measurement

1) While the ECM operates the leak detection pump, the purge VSV and vent valve turns off and a repeat 0.02 in. leak pressure measurement is performed.
2) The ECM compares the measured pressure with the pressure during EVAP leak check.
3) If the pressure during the EVAP leak check is below the measured value, the ECM determines that there is no leakage.
4) If the pressure during the EVAP leak check is above the measured value, the ECM determines that there is a small leak and stores DTC P0456 in its memory.



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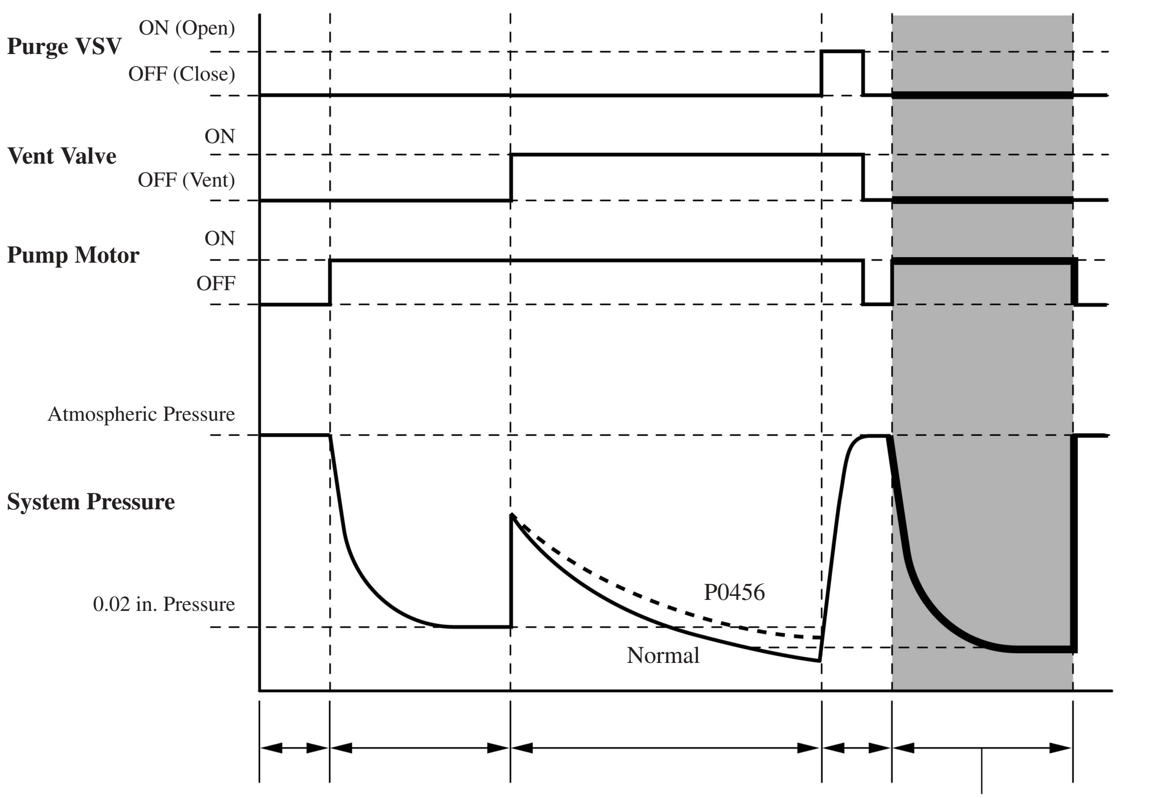

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12. Diagnosis

- When the ECM detects a malfunction, the ECM makes a diagnosis and memorizes the failed section. Furthermore, the MIL (Malfunction Indicator Lamp) in the combination meter illuminates or blinks to inform the driver.
- The ECM will also store the DTC (Diagnostic Trouble Code) of the malfunctions. The DTC can be accessed by using the hand-held tester.
- For details, see the 2007 Camry Repair Manual (Pub. No. RM0250U).

Service Tip

- The ECM of the ’07 Camry uses the CAN protocol for diagnostic communication. Therefore, a hand-held tester and a dedicated adapter [CAN VIM (Vehicle Interface Module)] are required for accessing diagnostic data. For details, see the 2007 Camry Repair Manual (Pub. No. RM0250U).
- To clear the DTC that is stored in the ECM, use a hand-held tester, disconnect the battery terminal or remove the EFI No.1 fuse and ETCS fuse for 1 minute or longer.

13. Fail-Safe

When the ECM detects a malfunction, the ECM stops or controls the engine according to the date already stored in the memory.

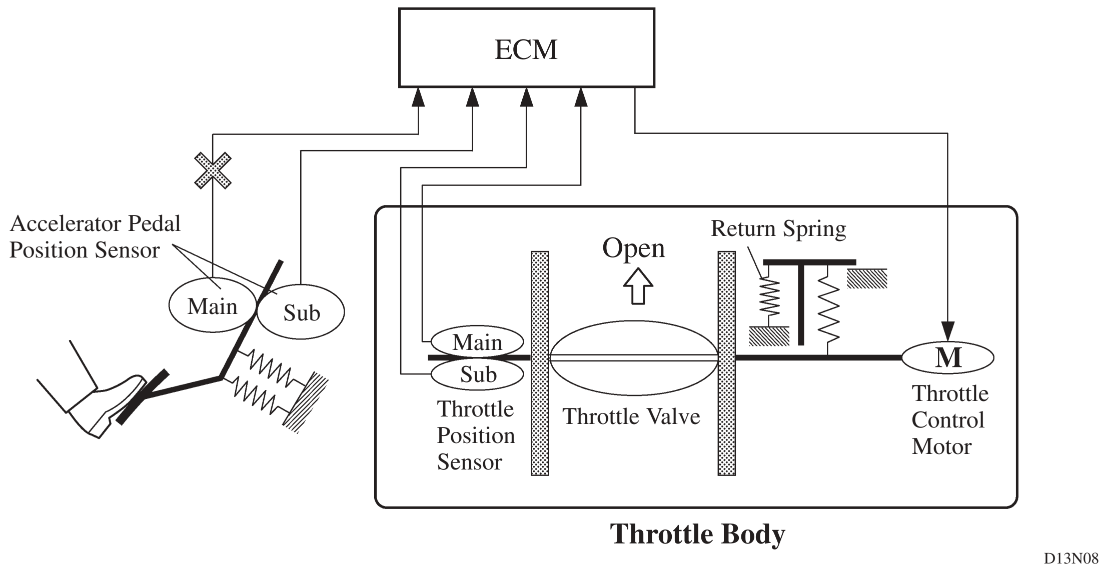
▶ **Fail-Safe Chart** ◀

DTC No.	Fail-Safe Operation	Fail-Safe Deactivation Conditions
P0031, P0032, P0037, P0038	The heater circuit in which an abnormality is detected is turned off.	Ignition switch OFF.
P0100, P0102, P0103	Ignition timing is calculated from an engine speed and a throttle angle.	Return to normal condition.
P0110, P0112, P0113	Intake air temp. is fixed at 20°C (68°F).	Return to normal condition.
P0115, P0117, P0118	Engine coolant temp. is fixed at 80°C (176°F).	Return to normal condition.
P0120, P0122, P0123, P0220, P0222, P0223, P2135	Fuel cut intermittently when idle.	Return to normal condition and ignition switch OFF.
P0121	Fuel cut intermittently when idle.	Return to normal condition and ignition switch OFF.
P0327, P0328	Max. timing retardation.	Ignition switch OFF.
P0351, P0352, P0353, P0354	Fuel cut.	Return to normal condition.
P2102, P2103	Fuel cut intermittently when idle.	Return to normal condition and ignition switch OFF.
P2111, P2112	Fuel cut intermittently when idle.	Return to normal condition and ignition switch OFF.
P2119	Fuel cut intermittently when idle.	Return to normal condition and ignition switch OFF.

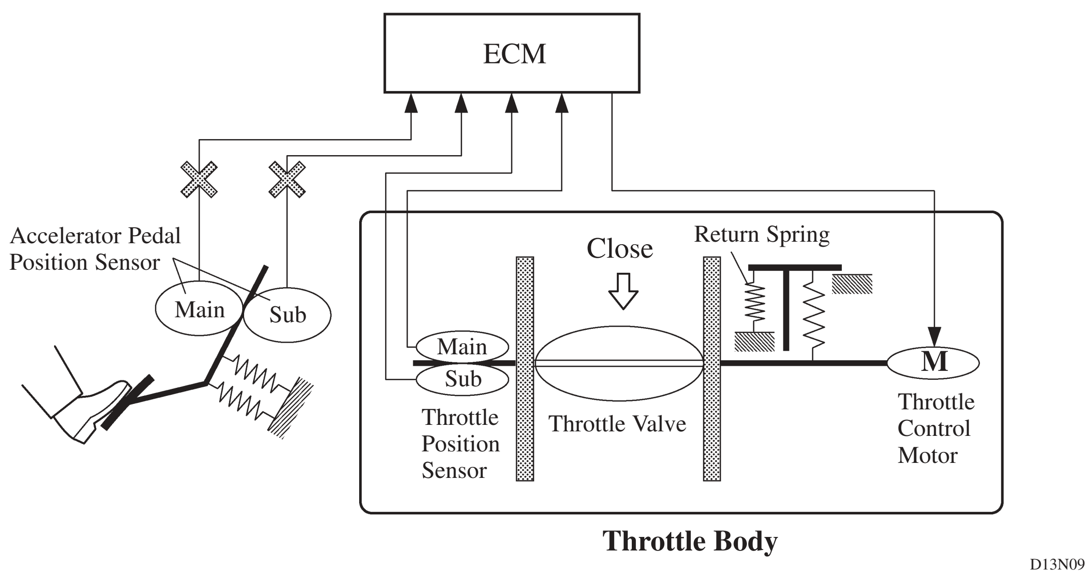
Fail-safe of Accelerator Pedal Position Sensor

The accelerator pedal position sensor comprises two (Main, Sub) sensor circuits.

- If a malfunction occurs in either of the sensor circuits, the ECM detects the abnormal signal voltage difference between these two sensor circuits and switches into the limp mode. In the limp mode, the remaining circuit is used to calculate the accelerator pedal opening, in order to operate the vehicle under limp mode control.



- If both circuits malfunction, the ECM detects the abnormal signal voltage from these two sensor circuits and discontinues the throttle control. At this time, the vehicle can be driven within its idling range.



Fail-safe of Throttle Position Sensor

The throttle position sensor comprises two (Main, Sub) sensor circuits.

- If a malfunction occurs in either of the sensor circuits, the ECM detects the abnormal signal voltage difference between these two sensor circuits, cuts off the current to the throttle control motor, and switches to the limp mode.
- Then, the force of the return spring causes the throttle valve to return and stay at the prescribed opening. At this time, the vehicle can be driven in limp mode while the engine output is regulated through the control of the fuel injection and ignition timing in accordance with the accelerator opening.
- The same control as above is effected if the ECM detects a malfunction in the throttle control motor system.